

CVT System Model Derivation

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Contents

1	Overview	4
1.1	Purpose of This Document	4
1.2	Scope	4
1.3	Expected Background Knowledge	4
1.4	Modeling Approach	4
2	Conceptual Overview of a Continuously Variable Transmission	5
2.1	Geometric Principle of a Belt-Sheave CVT	5
2.2	Continuous Ratio Versus Discrete Gearing	6
2.3	Traction-Based Torque Transmission	6
2.4	Actuation Mechanisms in the Present CVT	7
2.5	Full Drivetrain Architecture	9
3	Reference Material	10
3.1	Coordinate and Sign Conventions	10
3.1.1	Vehicle Longitudinal Coordinate and Incline Angle	10
3.1.2	Rotational Sign Convention	11
3.1.3	CVT Axial Direction	11
3.1.4	Radial and Tangential Directions	12
3.2	Symbol Summary	13
3.3	Base Units	13
3.4	Common Derived Units	14
3.5	Constants and Parameters	14
4	Modeling Assumptions	14
4.1	Material and Structural Idealizations	15
4.2	Contact and Friction Modeling	15
4.3	Geometric and Kinematic Idealizations	16
4.4	Dynamic and Inertial Modeling	16
4.5	Environmental and Operational Scope	17
5	Governing Principles	17
6	System State Definition	24
6.1	State Vector	25

6.2	Independent Dynamic Degrees of Freedom	25
6.3	Time Integration Perspective	25
7	Pulley Geometry	26
7.1	Conceptual Overview of Variable Geometry	26
7.2	Definition of Axial Shift Variable	27
7.3	Primary Outer Radius as a Function of Shift	28
7.4	Belt Length Constraint	29
7.5	Center-to-Center Distance Determination	31
7.6	Secondary Outer Radius as a Function of Shift	32
7.7	Wrap Angles as Functions of Shift	33
7.8	Effective Torque Radius	33
7.9	CVT Ratio Definition	34
7.10	Ratio Rate of Change	34
7.10.1	Start from the Ratio Definition	35
7.10.2	Differentiate R with Respect to s	35
7.10.3	Eliminating dr_s/ds Using the Belt-Length Constraint	35
7.10.4	Single Expression for dR/ds in Terms of Remaining Unknowns	36
7.10.5	Partial Derivatives of the Belt Constraint	36
7.10.6	Primary Radius Derivative	39
7.10.7	Final Expression for dR/ds and $\dot{R}$	39
7.11	Section Summary and Forward Connection	40
8	Pulley Axial Force Models	41
8.1	Ramp Geometry Parameterization	42
8.2	Primary Axial Force Model	44
8.3	Secondary Axial Force Model	46
8.4	Belt Centrifugal Force on the Wrap	49
8.5	Axial Shift Dynamics	53
8.6	Section Summary and Forward Outlook	55
9	Torque Transmission and Coupling Torque	56
9.1	System Definition and Torque Path	56
9.2	Rotational Equations of Motion	56
9.3	Engine Torque Model	57
9.4	Load Torque Model	57
9.5	Inertia Definitions	60
9.6	Ratio Kinematics	61
9.7	Torque Mapping Across the CVT	61
9.8	Eliminating Accelerations	62
9.9	Closed-Form Coupling Torque	63
9.10	Section Summary	63
10	Traction Limit and Slip Condition	63
10.1	Differential Belt Element and Tangential Force Balance	64
10.2	Integration Across the Wrap	66
10.3	Torque Capacity Inequality	66
10.4	Tangential Acceleration of a Belt Element on a Circular Wrap	67

10.5	Normal Force from Axial Clamping	69
10.6	Section Conclusion and Transition	70
11	Slip Regimes and Traction-Limited Coupling Torque	71
11.1	Working Backwards From the Goal	71
11.2	Slip Metric: How We Know We Are Slipping	71
11.3	Traction Capacity: How We Enter Slip	72
11.4	Coupling Torque Law: Stick When Feasible, Slip When Necessary	72
11.5	How Slip Reduces and Ends: Exiting the Slip Regime	73
11.6	Implementation Summary (Matches the Simulation Rule)	74
A	Parameter Tables	74
B	Belt Length Small-Angle Approximation	74
B.1	Standard Approximation	74
B.2	Error Analysis	74
C	Common Ramp Profile Designs	74
C.1	Linear Ramp	75
C.2	Quadratic Ramp	75
C.3	Piecewise-Linear Ramp	75
C.4	Custom Profiles from Lookup Tables	75
D	Numerical Methods Summary	75
D.1	Root Finding for Belt-Length Constraint	75
D.2	ODE Integration for Dynamic Simulation	75
E	Nomenclature and Symbol Reference	75

1 Overview

1.1 Purpose of This Document

This document presents a complete dynamic derivation of a continuously variable transmission (CVT) system model. The goal is to develop a physically consistent set of governing equations that describe:

- Rotational dynamics of the primary and secondary assemblies,
- Axial shift dynamics of the primary sheaves,
- Belt traction and torque transfer,
- Slip behavior under torque saturation,
- Coupled vehicle longitudinal response.

The resulting formulation provides a structured, state-based model suitable for numerical simulation and performance analysis.

1.2 Scope

This document focuses on the mechanical modeling of the CVT system. It includes force balances, torque transfer mechanisms, and dynamic coupling between rotational and axial motion.

The following are outside the scope of this document:

- Thermal modeling,
- Long-term wear or material degradation,
- Detailed tribological contact mechanics,
- Multi-body vehicle dynamics beyond one-dimensional motion.

All assumptions limiting model fidelity are formally defined in [Section 4](#).

1.3 Expected Background Knowledge

The reader is expected to be familiar with:

- Newton’s laws of motion,
- Basic rigid-body rotational dynamics,
- Linear spring models,
- Gear ratios and torque multiplication,
- Basic algebra and differential equations.

Prior familiarity with state-space modeling or numerical integration is not assumed. Relevant concepts are introduced where required.

1.4 Modeling Approach

The system is modeled using a minimal set of dynamic state variables, defined in [Section 6](#). All other quantities are derived algebraically from these states.

The derivation proceeds as follows:

- Governing physical laws are defined ([Section 5](#)),
- Kinematic relationships between pulley geometry and ratio are derived,

- Axial and radial force models are constructed,
- Belt traction limits are formulated,
- Rotational and axial equations of motion are assembled,
- Slip behavior is incorporated.

The final result is a coupled nonlinear dynamic system suitable for time-domain simulation.

2 Conceptual Overview of a Continuously Variable Transmission

This section introduces the fundamental operating principles of a continuously variable transmission (CVT) and then presents the specific CVT and drivetrain architecture considered in this work.

The goal is to establish clear mechanical intuition before developing the mathematical model in later sections.

2.1 Geometric Principle of a Belt-Sheave CVT

A belt-type CVT consists of two pulleys connected by a flexible belt. Each pulley is formed by two opposing conical faces (sheaves). The axial separation between these sheaves determines the effective radius at which the belt rides.

When the sheaves move closer together, the belt is forced outward to a larger radius. When the sheaves move apart, the belt settles at a smaller radius. Because the belt length remains approximately constant, an increase in effective radius on one pulley necessarily causes a corresponding decrease on the other.

As a result, the transmission ratio changes continuously as the pulley geometries change. Unlike a traditional gearbox, there are no discrete gear steps; the ratio evolves smoothly between its minimum and maximum values.

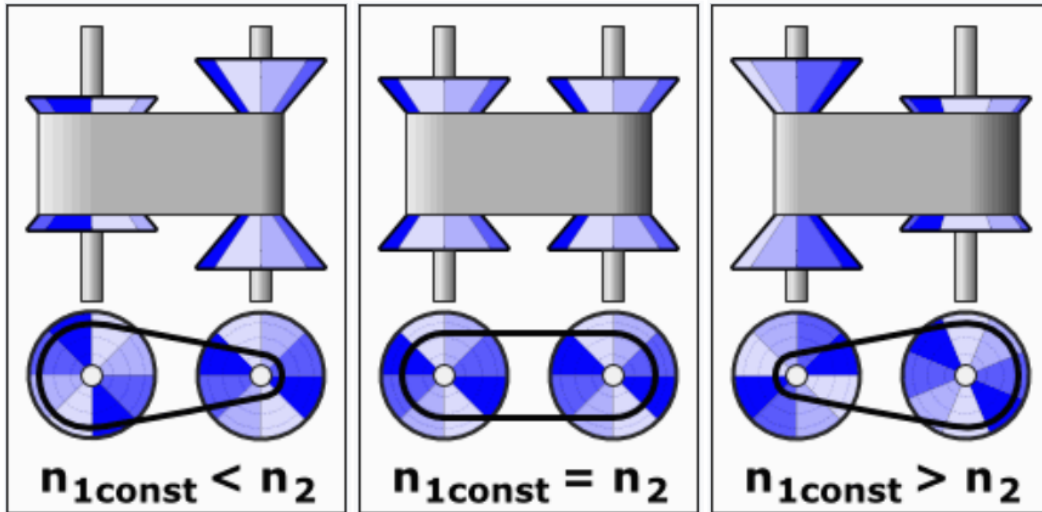


Figure 1: Conceptual illustration of a belt-sheave CVT at three different ratios. Changing the axial separation of the conical sheaves alters the effective belt radius, thereby modifying the transmission ratio continuously.

2.2 Continuous Ratio Versus Discrete Gearing

The defining feature of a CVT is that its transmission ratio can vary smoothly rather than in discrete steps. To understand why this matters, it is useful to first consider how a conventional transmission behaves.

In a stepped gearbox, each gear pair imposes a fixed relationship between engine speed and wheel speed. When vehicle speed increases beyond the useful range of one gear, the transmission must shift to another. This produces an abrupt change in engine speed for the same vehicle speed. As a result, the engine repeatedly accelerates and decelerates as the vehicle progresses through the gears.

A belt-sheave CVT operates differently. Because the effective pulley radii change continuously, the speed relationship between the engine and drivetrain can be adjusted smoothly at any instant. There are no fixed gear pairs that constrain the system to specific ratios. Instead, the ratio can evolve gradually as operating conditions change.

This continuous adjustability allows the engine to operate near a preferred speed while the vehicle accelerates. For example, if an engine produces its peak power within a narrow speed band, a CVT can maintain the engine near that region while vehicle speed increases steadily. Rather than stepping through gears, the transmission adapts its ratio to match the engine's most effective operating condition.

The result is smoother acceleration, improved power utilization, and greater flexibility in matching engine behavior to load demands.

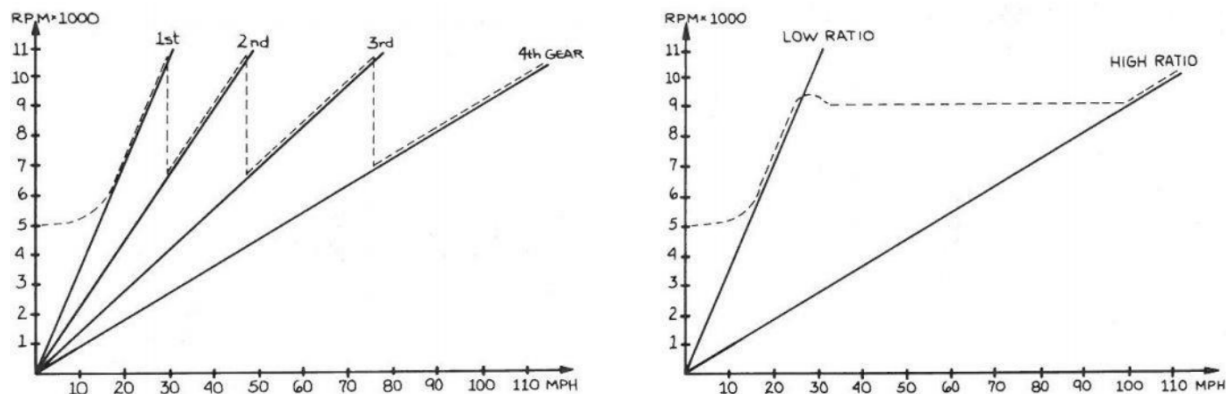


Figure 2: Illustrative comparison of engine speed versus vehicle speed. Left: stepped transmission showing discrete shifts. Right: continuously variable transmission showing smooth ratio evolution.

2.3 Traction-Based Torque Transmission

While the geometric adjustment of pulley radii determines the speed ratio, torque is transmitted through frictional interaction between the belt and the sheave faces.

When a pulley attempts to rotate, it applies a tangential force to the belt. For that force to be transmitted without relative motion, sufficient normal clamping force must press the belt against the conical surfaces. The greater the normal force, the greater the tangential force that can be supported before slip occurs.

This requirement arises directly from the nature of friction. The belt does not contain rigid gear

teeth that mechanically lock the pulleys together. Instead, torque transfer depends on traction generated by contact pressure and surface friction. If the transmitted torque exceeds the available traction capacity, the belt will begin to slip relative to the sheave faces.

Slip reduces efficiency, generates heat, and can accelerate component wear. For this reason, CVT designs incorporate mechanisms that actively regulate clamping force in response to operating conditions.

It is therefore useful to distinguish between two interacting aspects of CVT behavior:

- The **geometric ratio**, which determines relative speeds,
- The **traction capacity**, which determines how much torque can be transmitted without slip.

Both must be considered to fully understand performance.

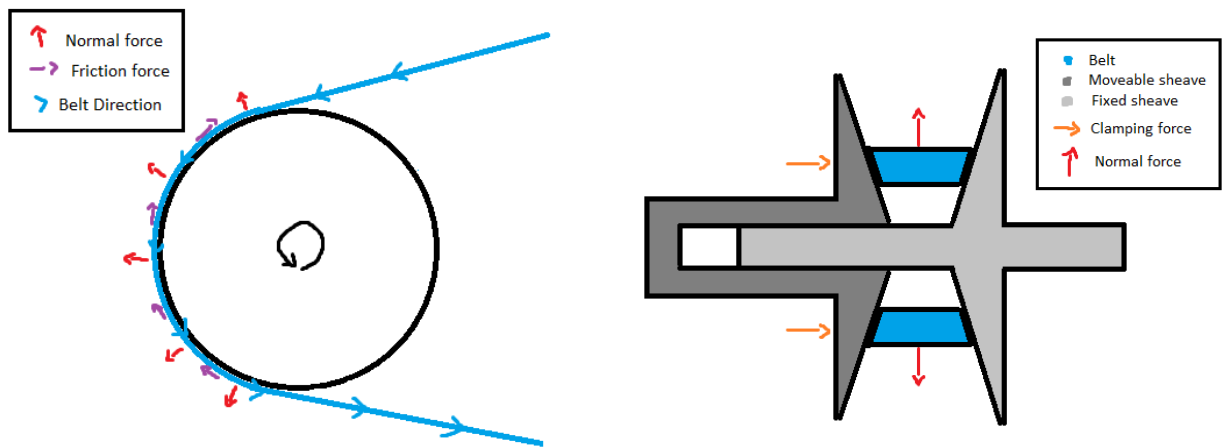


Figure 3: Belt-pulley contact mechanics and force transmission. Left: belt wrapped around a pulley showing normal forces (pressing belt against sheave surface) and tangential friction forces (transmitting torque through traction). Right: cross-sectional view of a CVT pulley assembly showing how axial clamping force between opposing sheaves generates the normal force required for friction and torque transmission.

2.4 Actuation Mechanisms in the Present CVT

While the geometric principles described above apply to all belt-type CVTs, the manner in which axial motion and clamping forces are generated depends on the specific hardware implementation.

The CVT considered in this work employs:

- A **centrifugal flyweight and ramp mechanism** on the primary pulley,
- A **torque-feedback helix mechanism** on the secondary pulley,
- Compression and torsional springs that regulate axial clamping forces.

On the primary side, rotating flyweights move outward under centrifugal effects. Through a ramped interface, this outward motion generates axial force, which adjusts the sheave separation and alters the effective radius.

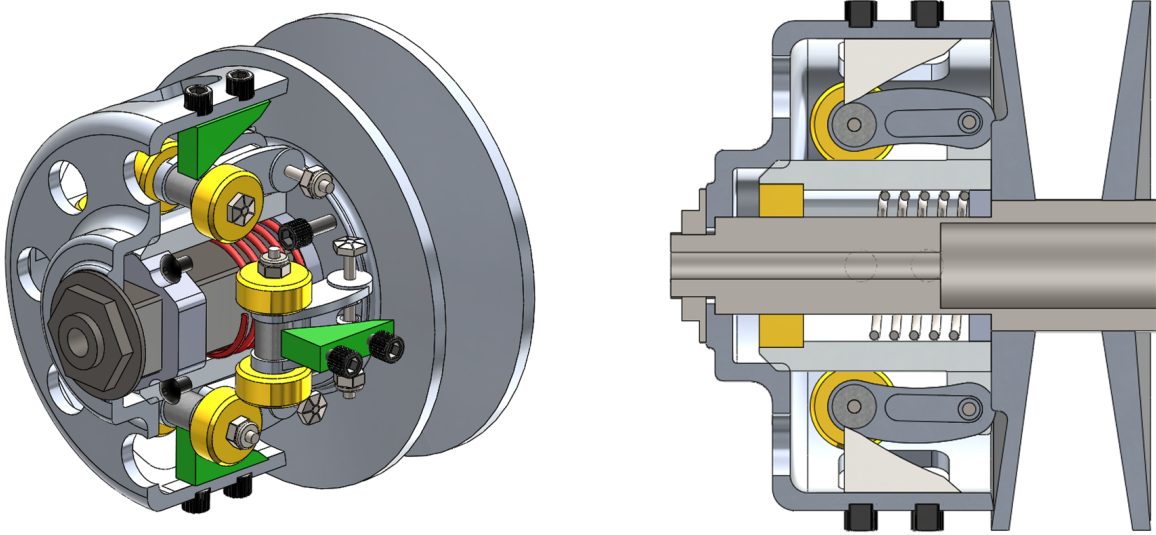


Figure 4: Primary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing centrifugal flyweights (yellow), ramps (green), and compression spring (red). As engine speed increases, centrifugal force drives the flyweights outward along the ramps, generating axial clamping force.

On the secondary side, transmitted torque produces a rotational reaction within a helical interface. This interaction generates axial force that increases clamping in response to torque demand. Springs supplement this mechanism to regulate baseline clamping and system stability.

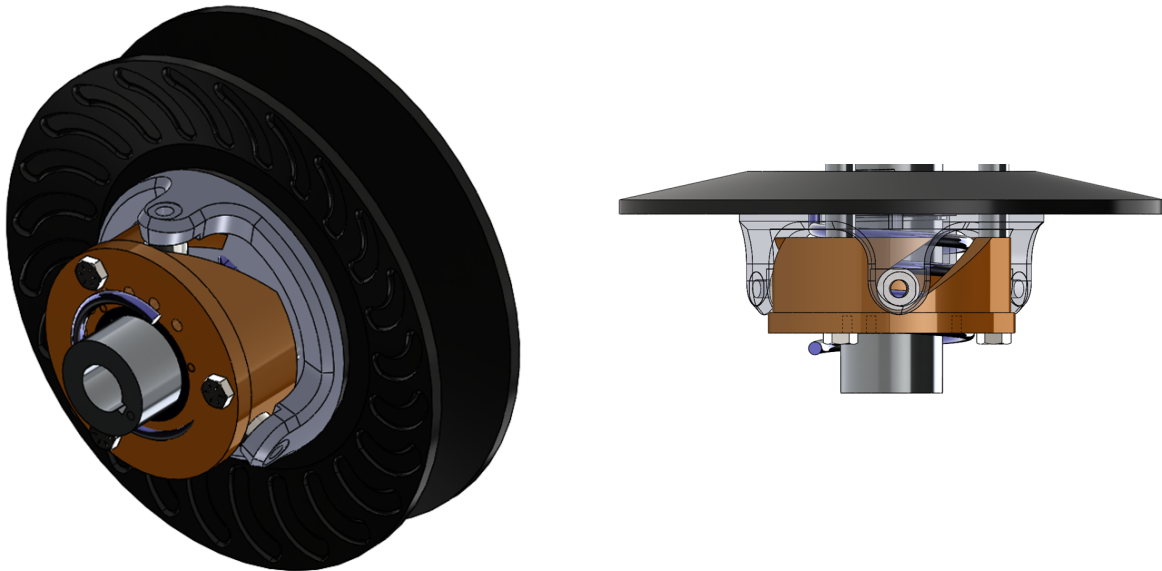


Figure 5: Secondary CVT pulley assembly. Left: exterior view. Right: cross-sectional view showing torque-feedback helix mechanism (bronze) and torsional spring (black). Transmitted torque generates a rotational reaction in the helix, which produces axial clamping force proportional to load.

Although these mechanisms are specific to the hardware used in this system, the underlying principles of variable effective radius and traction-based torque transmission remain common to belt-driven CVTs.

2.5 Full Drivetrain Architecture

The CVT does not operate in isolation. In the system considered here, the drivetrain consists of:

- An internal combustion engine,
- The primary CVT pulley,
- A belt connecting to the secondary CVT pulley,
- A fixed-ratio gearbox,
- Final drive components and driven wheels,
- Tire-ground interaction generating tractive forces.

Engine torque enters the primary pulley, is transmitted through the belt to the secondary pulley, passes through a constant gear reduction, and ultimately produces tractive force at the ground.

The interaction between engine dynamics, CVT ratio evolution, drivetrain inertia, and external load forces governs overall vehicle performance.

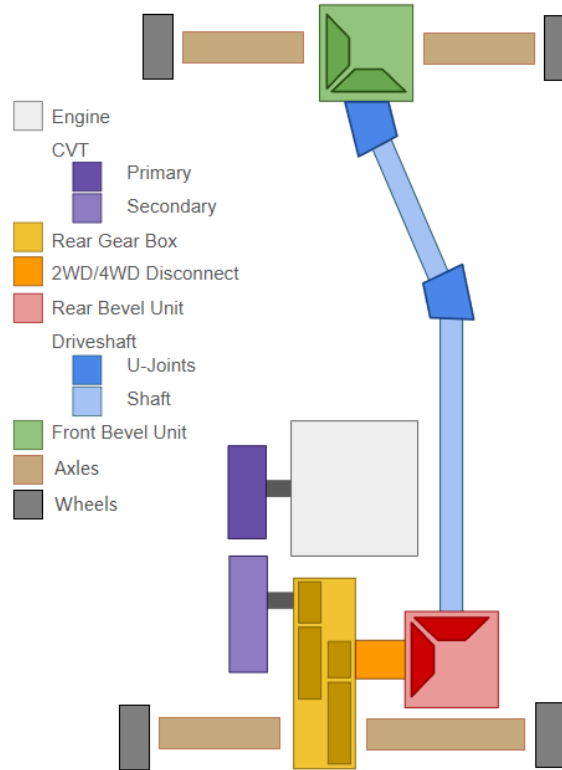


Figure 6: Overview of the complete drivetrain architecture considered in this work.

3 Reference Material

This section records any information for easy reference.

3.1 Coordinate and Sign Conventions

A consistent set of coordinate directions and sign conventions is adopted throughout this document. All equations follow these conventions.

3.1.1 Vehicle Longitudinal Coordinate and Incline Angle

The vehicle is modeled as moving along a single straight line aligned with the road surface.

The positive longitudinal direction, denoted $+x$, is defined as the forward direction of travel along the surface of the road.

The road may be inclined relative to horizontal. The incline angle is denoted by α and is measured between the road surface and the horizontal reference line. An incline with $\alpha > 0$ corresponds to the vehicle facing uphill in the positive x direction.

Importantly, the $+x$ axis is *tangent to the road surface* and rotates with the road inclination. The $+x$ direction is not fixed in space but rather follows the path of the road, changing orientation as the incline angle α changes.

Because the $+x$ axis remains parallel to the road surface regardless of incline, the coordinate system rotates through angle α relative to a horizontal reference frame.

A vehicle with velocity $v > 0$ is therefore moving forward along the road in the $+x$ direction. If $v < 0$, the vehicle is moving in the opposite direction.

These definitions establish the geometric orientation of vehicle motion and the road surface.

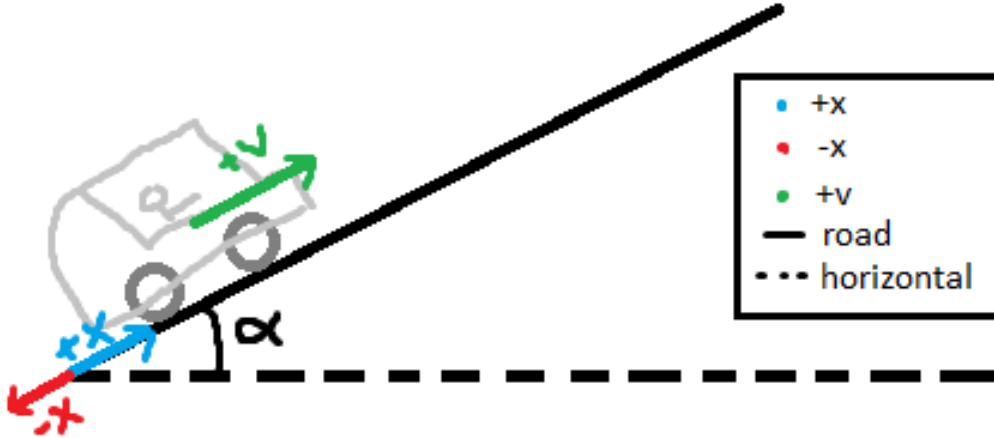


Figure 7: Definition of the longitudinal coordinate and incline angle. The positive x direction follows the road surface. Velocity v is positive when directed uphill along $+x$. The incline angle α is measured between the road and the horizontal.

3.1.2 Rotational Sign Convention

All rotating components in the drivetrain follow a consistent angular sign convention.

The primary pulley, secondary pulley, and engine crankshaft are each assigned a positive direction of rotation. This positive rotational direction is defined such that it produces forward belt motion and, consequently, forward vehicle motion along the $+x$ direction defined previously.

Angular velocities are therefore interpreted as follows:

- $\omega_p > 0$ corresponds to the primary pulley rotating in the direction that drives the belt forward.
- $\omega_s > 0$ corresponds to the secondary pulley rotating in the direction consistent with forward belt travel.
- Engine angular velocity is positive when rotating in the direction that drives the primary pulley forward.

Torque follows the same convention. A torque is considered positive if it acts in the defined positive direction of rotation and tends to increase the corresponding angular velocity. A torque acting opposite to the defined rotational direction is negative.

These definitions establish a consistent orientation for all rotational quantities in the drivetrain.

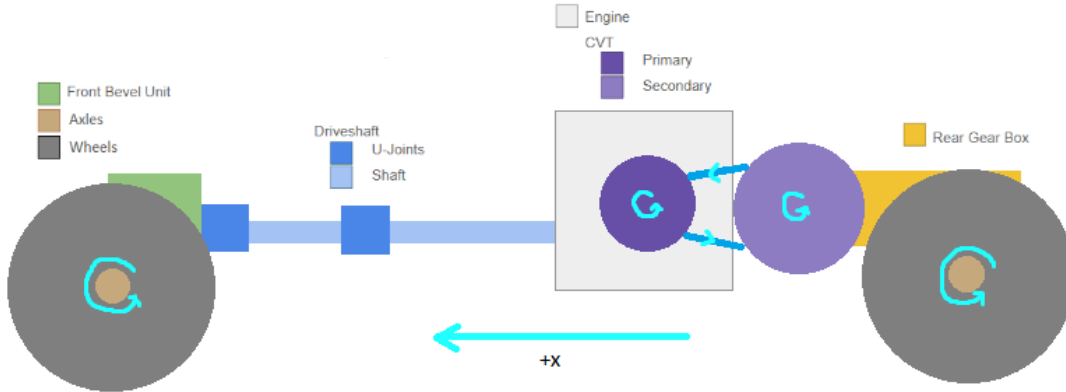


Figure 8: Rotational sign convention for the drivetrain. Positive angular velocities (ω_p , ω_s) correspond to rotation that drives the belt in the $+x$ direction, producing forward vehicle motion. Positive torque acts in the same rotational direction.

3.1.3 CVT Axial Direction

The axial direction of the primary pulley is defined along the axis of its shaft. This direction is parallel to the pulley centerline and perpendicular to the belt's plane of rotation.

The primary pulley is initially in an open configuration, meaning there is a finite gap between the fixed and movable sheaves. The axial coordinate s does not represent the total gap distance. Instead, it measures how much the movable sheave has translated from this initial open position.

A positive axial displacement, $s > 0$, corresponds to the movable sheave moving toward the fixed sheave, thereby reducing the gap. In other words, increasing s represents additional closure relative to the initial open state.

As the sheaves move closer together, the belt is forced outward along the conical faces, increasing the effective belt radius on the primary pulley. Thus, increasing s corresponds to increasing primary effective radius and produces an upshift.

A change in s also affects the secondary pulley. Because the belt length remains constrained, motion of the primary sheaves is coupled to motion of the secondary sheaves. In general terms, an increase in s on the primary corresponds to the secondary sheaves moving apart. The precise relationship between these motions depends on belt geometry and will be established later; here it is sufficient to recognize that the two axial motions are linked.

Axial velocity follows the same sign convention: $\dot{s} > 0$ corresponds to the movable sheave translating in the defined positive axial direction.

These definitions establish the orientation and meaning of axial motion for the CVT.

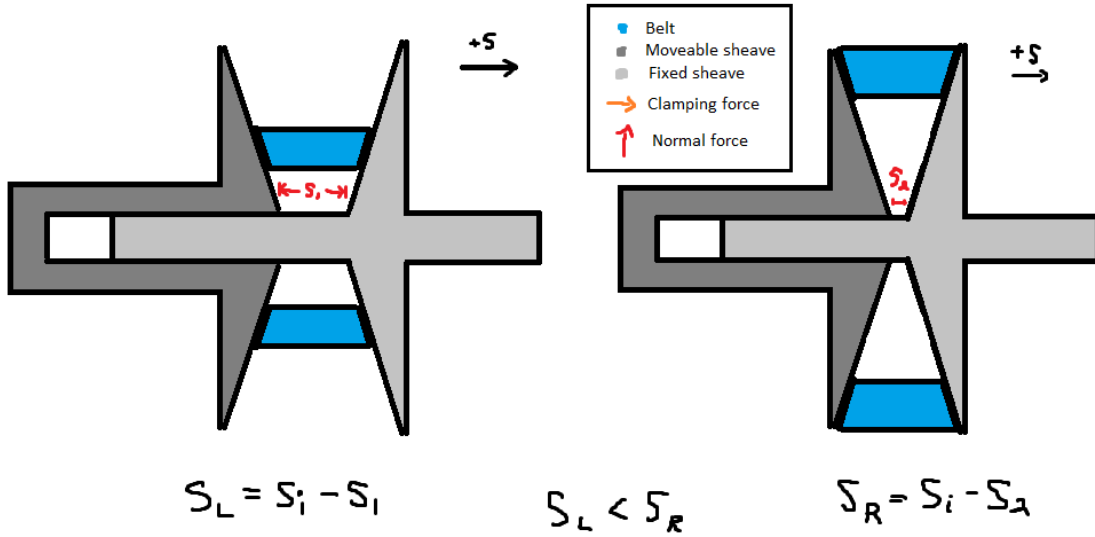


Figure 9: Definition of the primary axial coordinate. The system begins in an open configuration. The axial displacement s measures how much the movable sheave has closed from this initial position. Increasing s increases primary effective radius and is coupled to secondary axial motion.

3.1.4 Radial and Tangential Directions

Within the plane of pulley rotation, directions are defined using the standard polar (radial–tangential) convention commonly used in rotational mechanics.

The radial direction is defined perpendicular to the shaft axis and measured outward from the center of rotation. At any point on the pulley, the radial direction points directly away from the center toward the belt contact surface.

A radial force is defined as positive when it acts outward from the center of the pulley. Thus, $F_r > 0$ corresponds to a force directed away from the center.

The tangential direction lies in the plane of rotation and is perpendicular to the radial direction at each point. The positive tangential direction is defined to be consistent with the previously defined positive angular rotation. In other words, positive tangential direction corresponds to the direction of motion produced by positive angular velocity defined in [Section 3.1.2](#).

Together, the radial and tangential directions form the standard local coordinate basis for forces and motion in the rotating pulley.

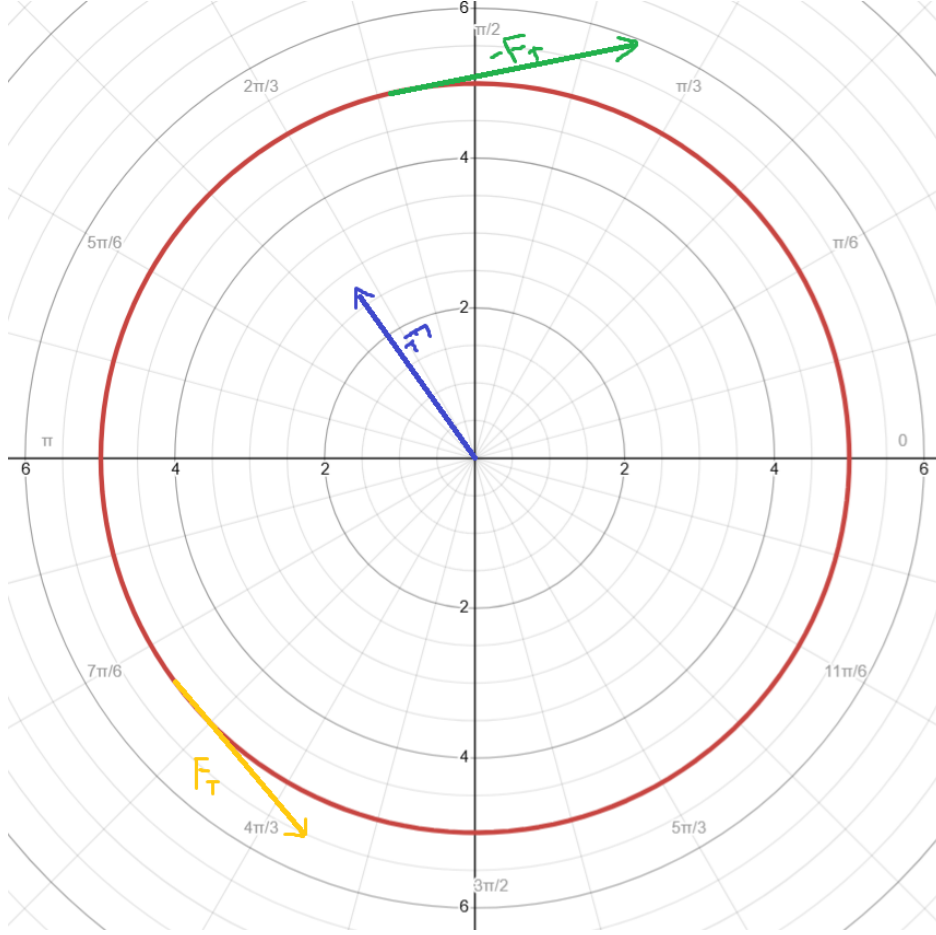


Figure 10: Definition of radial and tangential directions in the plane of rotation. The radial direction points outward from the center. The tangential direction is perpendicular to the radial direction and aligned with positive rotation.

3.2 Symbol Summary

Symbol	Meaning
v	vehicle speed
ω_e	engine angular speed
s	primary axial shift position
$\dot{s}$	primary axial shift velocity
R	CVT ratio (define direction explicitly)

Table 1: Core symbols used throughout the derivation.

3.3 Base Units

All quantities in this document are expressed in SI units unless otherwise stated.

Quantity	Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	s
Plane angle	radian	rad

Table 2: SI base units used throughout the model.

3.4 Common Derived Units

Quantity	Unit	Symbol
Velocity	meter per second	m/s
Acceleration	meter per second squared	m/s ²
Angular velocity	radian per second	rad/s
Angular acceleration	radian per second squared	rad/s ²
Force	newton (kg m/s ²)	N
Torque	newton meter (kg m ² /s ²)	N m
Moment of inertia	kilogram meter squared	kg m ²
Power	watt (kg m ² /s ³)	W

Table 3: Common derived SI units used in the CVT model.

3.5 Constants and Parameters

All physical constants, geometric parameters, and calibrated values used in the model are listed below.

Symbol	Name	Value	Unit	Description
m	Vehicle mass	–	kg	Total vehicle mass
r_w	Wheel radius	–	m	Effective rolling radius
G	Final drive ratio	–	–	Secondary-to-wheel speed ratio
I_p	Primary inertia	–	kg m ²	Lumped primary-side inertia
I_s	Secondary inertia	–	kg m ²	Lumped secondary-side inertia
k_p	Primary spring constant	–	N/m	Primary compression spring
k_s	Secondary spring constant	–	N/m	Secondary compression spring
μ	Friction coefficient	–	–	Belt-pulley friction coefficient
β	Wrap angle	–	rad	Belt wrap angle

Table 4: Model constants and parameters.

4 Modeling Assumptions

The following assumptions define the modeling scope and physical idealizations used in the CVT dynamic formulation. These assumptions establish the limits within which the derived equations remain valid.

4.1 Material and Structural Idealizations

Assumption 1 (Constant Material Properties)

Material properties of structural components are treated as constant. No temperature-dependent material variation is modeled. Thermal effects are not solved as part of this system.

Assumption 2 (Rigid Structural Components)

All structural components (pulleys, shafts, housing) are modeled as rigid bodies. Elastic deformation under operational loading is neglected. The belt is assumed axially rigid (no longitudinal stretch), while still conforming geometrically around pulleys.

Assumption 3 (Negligible Wear and Surface Evolution)

Component wear, belt degradation, and pulley surface evolution are neglected. The model represents short-term operational behavior.

4.2 Contact and Friction Modeling

Assumption 4 (Idealized Belt–Pulley Friction Law)

Belt traction is modeled using a simplified friction law (e.g., Coulomb friction with a capstan formulation and an explicit stick–slip transition rule). Microscopic contact mechanics, rubber hysteresis, lubrication films, and detailed Stribeck effects are not explicitly modeled.

Assumption 5 (Constant Effective Friction Coefficient)

The belt–pulley coefficient of friction is treated as constant. It does not evolve with temperature, wear, or surface conditioning.

Assumption 6 (Uniform Contact Distribution)

Contact pressure and traction over the pulley wrap angle are treated as effectively uniform or averaged. Localized edge loading and non-uniform stress distributions are neglected.

4.3 Geometric and Kinematic Idealizations

Assumption 7 (Perfect Axis Alignment)

Primary and secondary pulley axes are assumed perfectly aligned. Misalignment, shaft eccentricity, and geometric runout are neglected.

Assumption 8 (Constant Belt Length)

The belt length is treated as constant. Permanent creep and progressive elongation are not modeled. Effective pulley radii are determined solely from geometry and belt-length constraints.

4.4 Dynamic and Inertial Modeling

Assumption 9 (Lumped Inertia Representation)

Rotational inertias of the engine, pulleys, drivetrain, and wheels are represented as lumped equivalent inertias. Distributed shaft flex and torsional compliance are neglected.

Assumption 10 (Negligible Vibrational Effects)

High-frequency vibrational effects, structural resonances, and drivetrain oscillations are neglected. The model captures dominant rigid-body dynamics only.

Assumption 11 (Negligible Parasitic Friction)

Frictional losses in non-critical components (e.g., bearings, seals, auxiliary shafts) are neglected. Only friction directly relevant to belt traction and CVT torque transfer is modeled.

Assumption 12 (Ideal CVT Power Transmission)

Power transmission through the CVT is modeled as ideal (i.e., no internal dissipation). Although belt slip is explicitly modeled and affects torque capacity and shift dynamics, the associated power loss P_{loss} is neglected.

4.5 Environmental and Operational Scope

Assumption 13 (Negligible Gravitational Influence on CVT Internals)

Gravitational forces acting on CVT internal components are negligible relative to centrifugal and contact forces at operating speeds.

Assumption 14 (Full-Throttle Engine Model)

The engine is modeled using a prescribed full-throttle torque curve. Load feedback modifies engine speed dynamically, but part-throttle dynamics and throttle transients are not explicitly modeled.

5 Governing Principles

Theory 1: Newton's Second Law (Translation)

Equation

$$\sum F = ma$$

Description

- $\sum F$ is the net force (N)
- m is mass (kg)
- a is acceleration (m/s²)

Source

Classical mechanics (textbook / lecture notes).

Derivation

From the definition of acceleration and empirical laws of motion; used here as a governing relationship.

Theory 2: Newton's Second Law (Rotation)

Equation

$$\sum \tau = I\alpha$$

Description

- $\sum \tau$ is net torque (N m)
- I is moment of inertia (kg m²)
- α is angular acceleration (rad/s²)

Source

Classical rigid body dynamics (textbook / lecture notes).

Derivation

Standard rigid-body rotational dynamics relationship.

Theory 3: Torque Definition

Equation

$$\tau = Fr$$

Description

- τ is torque (N m)
- F is tangential force (N)
- r is moment arm radius (m)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Defines torque as the product of tangential force and moment arm. Equivalently, $F = \tau/r$ gives the tangential force from a known torque.

Theory 4: Kinetic Energy (Translational)

Equation

$$T = \frac{1}{2}mv^2$$

Description

- T is kinetic energy (J)
- m is mass (kg)
- v is velocity (m/s)

Source

Classical mechanics (textbook / lecture notes).

Derivation

Translational kinetic energy of a point mass or rigid body moving with velocity v .

Theory 5: Centrifugal Force

Equation

$$F_c = m\omega^2 r$$

Description

- F_c is centrifugal force (N)
- m is mass (kg)
- ω is angular velocity (rad/s)
- r is radius from the axis of rotation (m)

Source

[Bogna Szyk and Steven Wooding \(2024\)](#).

Derivation

Derived from circular motion kinematics where $a_c = \omega^2 r$, then applying $F = ma$.

Theory 6: Hooke's Law (Compressional)

Equation

$$F = k\Delta x$$

Description

- F is force (N)
- k is spring constant (N/m)
- Δx is displacement (m)

Source

[Encyclopaedia Britannica](#) (2024).

Derivation

Linear spring constitutive relationship in the elastic regime.

Theory 7: Hooke's Law (Torsional)

Equation

$$F = \frac{k\Delta\theta}{r}$$

Description

- F is force (N)
- k is torsional spring constant (N m/rad)
- $\Delta\theta$ is angular deflection (rad)
- r is torque arm radius (m)

Source

[Dr. Templeman](#) (2024).

Derivation

Starting from $\tau = k\Delta\theta$ and $\tau = Fr$, then $F = \frac{k\Delta\theta}{r}$.

Theory 8: Engine Torque from Power

Equation

$$\tau = \frac{P}{\omega}$$

Description

- τ is torque (N m)
- P is power (W)
- ω is angular velocity (rad/s)

Source

2024 Power Test, LLC (2024).

Derivation

From the rotational power relationship $P = \tau\omega$.

Theory 9: Gearing (Torque Scaling)

Equation

$$\tau_{\text{out}} = \tau_{\text{in}} R_{\text{gear}}$$

Description

- τ_{out} is output torque (N m)
- τ_{in} is input torque (N m)
- R_{gear} is gear reduction ratio (unitless)

Source

Brain (2023).

Derivation

Ideal gear relationship assuming no losses; power balance implies inverse scaling of speed.

Theory 10: Coulomb Friction

Equation

$$F_f = \mu_s F_n$$

Description

- F_f is friction force (N)
- μ_s is coefficient of static friction (unitless)
- F_n is normal force (N)

Source

[Wikipedia contributors](#) (2024).

Derivation

Coulomb friction model for impending slip.

Theory 11: Capstan Equation

Equation

$$T_1 = T_2 e^{\mu\theta}$$

Description

- T_1 is tight-side belt tension (N)
- T_2 is slack-side belt tension (N)
- μ is belt–pulley friction coefficient (unitless)
- θ is wrap angle (rad)

Source

[Hackaday](#) (2021).

Derivation

Differential force balance on an infinitesimal belt element on a pulley yields $\frac{dT}{T} = \mu \, d\theta$, then integrate over wrap.

Theory 12: Density

Equation

$$\rho = \frac{m}{V}$$

Description

- ρ is density (kg/m³)
- m is mass (kg)
- V is volume (m³)

Source

[The Editors of Encyclopaedia Britannica \(2024\)](#).

Derivation

Definition of density.

Theory 13: Air Resistance (Quadratic Drag)

Equation

$$F_d = \frac{1}{2}\rho v^2 C_d A$$

Description

- F_d is drag force (N)
- ρ is fluid density (kg/m³)
- v is object speed relative to fluid (m/s)
- C_d is drag coefficient (unitless)
- A is reference area (m²)

Source

[SoftSchools \(2020\)](#).

Derivation

Standard empirical/derived quadratic drag model; applicable at moderate to high Reynolds number.

Theory 14: Rolling Resistance

Equation

$$F_r = C_r N$$

Description

- F_r is rolling resistance force (N)
- C_r is coefficient of rolling resistance (unitless)
- N is normal force (N)

Source

?.

Derivation

Rolling resistance arises from deformation of contact surfaces (tire and road). For a vehicle on level ground, $N = mg$ where m is vehicle mass and g is gravitational acceleration.

Theory 15: Gravitational Force

Equation

$$F_g = mg$$

Description

- F_g is gravitational force (N)
- m is mass (kg)
- g is gravitational acceleration (m/s^2)

Source

?.

Derivation

Near Earth's surface, $g \approx 9.81 \text{ m/s}^2$. On an incline with angle θ , the component parallel to the surface is $F_g \sin \theta$.

6 System State Definition

The dynamic behavior of the CVT system is described using a set of *state variables*. A state variable is a quantity whose future evolution is determined by its current value and the governing differential equations of the system.

In numerical simulation, state variables are the quantities that are explicitly integrated forward

in time. All other model quantities — such as torque transfer, transmission ratio, belt forces, and contact conditions — are computed algebraically from these states.

6.1 State Vector

State Vector	(6.1)
$\mathbf{x}(t) = \begin{bmatrix} \omega_p \\ \omega_s \\ s \\ \dot{s} \end{bmatrix}$	

where:

- ω_p is the angular speed of the primary pulley (rad/s),
- ω_s is the angular speed of the secondary pulley (rad/s),
- s is the axial shift position of the primary sheave (m),
- $\dot{s}$ is the axial shift velocity (m/s).

6.2 Independent Dynamic Degrees of Freedom

These four quantities represent the independent dynamic motions of the CVT system.

The primary pulley can rotate independently under engine input and belt interaction. Its angular speed ω_p therefore represents one rotational degree of freedom.

The secondary pulley also rotates and interacts with both the belt and the external drivetrain load. Its angular speed ω_s represents a second independent rotational degree of freedom.

In addition to rotation, the CVT includes axial motion of the primary sheave. This axial displacement s determines the effective pulley geometry and therefore the transmission ratio. The axial motion is mechanically independent of pure rotation and must be modeled explicitly.

Because axial motion follows second-order dynamics, both the axial position s and its rate $\dot{s}$ are required to fully describe its evolution.

Together, these variables capture the complete rotational and axial motion of the system. Removing any one of them would eliminate an independent dynamic degree of freedom from the model.

6.3 Time Integration Perspective

During numerical simulation, the state vector is advanced forward in time.

At each timestep Δt :

- The angular speeds ω_p and ω_s are updated according to their angular accelerations.
- The axial position s is updated using its current velocity $\dot{s}$.
- The axial velocity $\dot{s}$ is updated according to the axial acceleration.

The angular and axial accelerations are determined from the governing equations derived in subsequent sections. Once the state vector has been updated, all other quantities in the model are computed directly from the current values of ω_p , ω_s , s , and $\dot{s}$.

7 Pulley Geometry

This section establishes the geometric relationships that govern how axial shift modifies the effective pulley radii and therefore the transmission ratio.

The analysis presented here is purely kinematic. No force balances or dynamic effects are considered. The goal is to describe how pulley geometry alone determines belt position and ratio evolution.

The objective is to derive explicit expressions for:

- Primary outer radius as a function of shift,
- Secondary outer radius as a function of shift,
- Center-to-center distance,
- CVT ratio and its rate of change.

7.1 Conceptual Overview of Variable Geometry

As introduced in [Section 2](#), each pulley consists of two opposing conical sheaves. One sheave is fixed axially, while the other translates along the shaft.

The axial separation between the sheaves determines where the belt contacts the conical faces. When the movable sheave translates toward the fixed sheave, the gap decreases and the belt is forced outward, increasing the effective pulley radius. When the sheaves separate, the belt settles deeper between them, decreasing the effective radius.

It is important to distinguish between axial displacement and effective radius. Axial motion changes the gap between sheaves; the effective radius changes as a geometric consequence of how the belt sits within that gap. The relationship between these quantities will be derived explicitly in the following subsection.

Because the belt length remains constrained, the effective radii of the primary and secondary pulleys cannot vary independently. An increase in the primary effective radius must be accompanied by a corresponding decrease in the secondary effective radius. This geometric coupling is fundamental to CVT operation and forms the basis of the ratio model derived below.

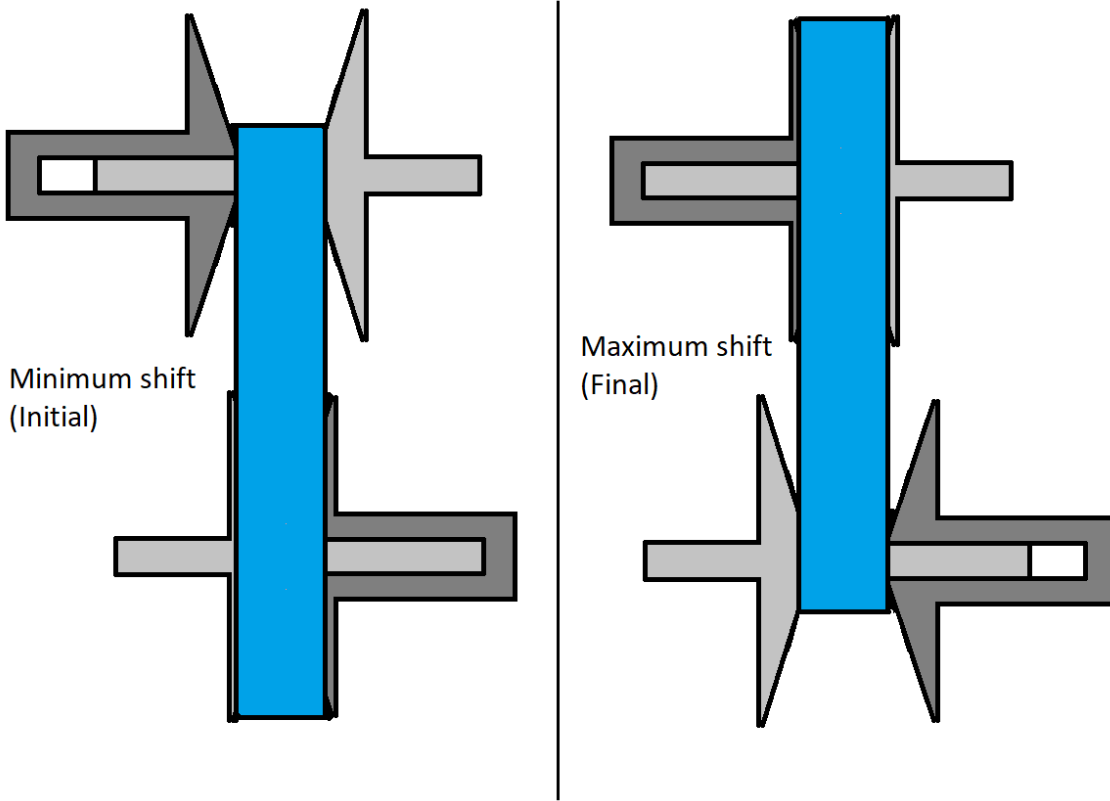


Figure 11: Conceptual illustration of variable pulley geometry. Changes in axial separation alter effective radii. Because belt length is constrained, an increase in primary effective radius corresponds to a decrease in secondary effective radius.

7.2 Definition of Axial Shift Variable

Let s denote the axial displacement of the movable primary sheave, defined according to the sign convention in [Section 3.1.3](#).

At $s = 0$, the primary pulley is in its fully open configuration. The sheaves are at their maximum axial separation, and the belt rests at its minimum effective radius.

The shift coordinate s measures the amount of closure relative to this initial configuration. Increasing s corresponds to the movable sheave translating toward the fixed sheave, reducing the axial gap.

Deadzone and Travel Limits In practice, the initial sheave gap at $s = 0$ is larger than the belt width. As a result, small axial motion does not immediately alter the belt's contact position on the conical faces.

The interval

$$0 \leq s < s_{dz}$$

defines this *deadzone*. Within this region, the movable sheave translates axially but does not yet sufficiently compress the belt to change its effective contact radius. The belt remains at the same radial position despite the reduction in gap.

Once the sheave gap becomes small enough to fully engage the belt, further axial motion alters the belt position. For

$$s_{dz} \leq s \leq s_{\max},$$

additional closure forces the belt outward along the conical faces, producing a proportional change in primary effective radius.

The shift coordinate is therefore bounded as

$$0 \leq s \leq s_{\max}.$$

The upper limit $s_{\max}$ corresponds to the physical travel limit of the sheave assembly. At this position, the sheaves are fully closed and mechanically contact one another, forming a hard geometric boundary beyond which further axial motion is not possible.

7.3 Primary Outer Radius as a Function of Shift

When the primary sheaves close beyond the deadzone, the belt is forced outward along the conical faces. The relationship between axial closure and radial expansion follows directly from the cone geometry.

As shown in [Figure 12](#), the sheave face forms a straight line in cross-section. The cone half-angle β is defined as the angle between the sheave face and the *radial direction* (As defined in [Section 3.1.4](#)).

For a small axial displacement Δs beyond the deadzone, the total closure is shared symmetrically between the two opposing sheaves, so each sheave contributes $\Delta s/2$.

From the right triangle formed by the sheave face,

$$\tan \beta = \frac{\Delta s/2}{\Delta r},$$

which gives

$$\Delta r = \frac{\Delta s}{2 \tan \beta}.$$

Thus, the cone angle determines how axial motion is converted into radial growth: smaller β produces greater radial change for the same axial displacement.

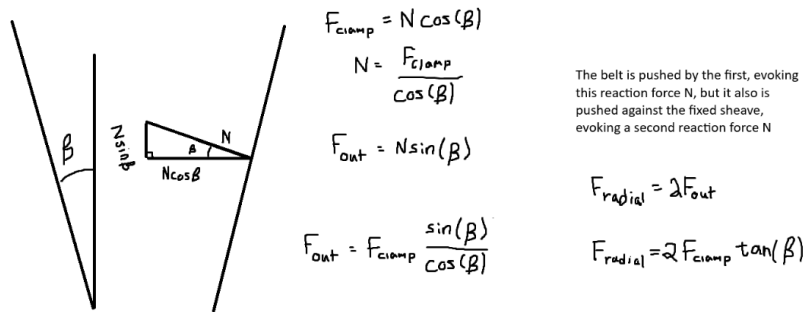


Figure 12: Axial-radial relationship for opposing conical sheaves. The angle β is defined between the sheave face and the radial direction.

Piecewise Radius Model Let $r_{p,\min}$ denote the minimum primary outer radius, corresponding to the fully open configuration ($s = 0$).

Within the deadzone, axial motion does not alter belt position. Once $s \geq s_{dz}$, the geometric relation above applies, yielding

Primary Outer Radius

(7.1)

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\min}, & 0 \leq s < s_{dz}, \\ r_{p,\min} + \frac{s - s_{dz}}{2 \tan \beta}, & s_{dz} \leq s \leq s_{\max}. \end{cases}$$

7.4 Belt Length Constraint

In the previous subsection, the primary effective radius was expressed as a function of axial shift. The secondary radius, however, cannot be chosen independently. The belt has a fixed total length, and therefore any change in primary radius must be accompanied by a corresponding geometric adjustment of the secondary radius.

This section derives the geometric compatibility constraint imposed by constant belt length. By expressing the total belt length in terms of the primary radius r_p , secondary radius r_s , and center-to-center distance C , we obtain an implicit relationship that couples the two pulley radii.

The derivation follows directly from the belt path geometry shown in [Figure 13](#).

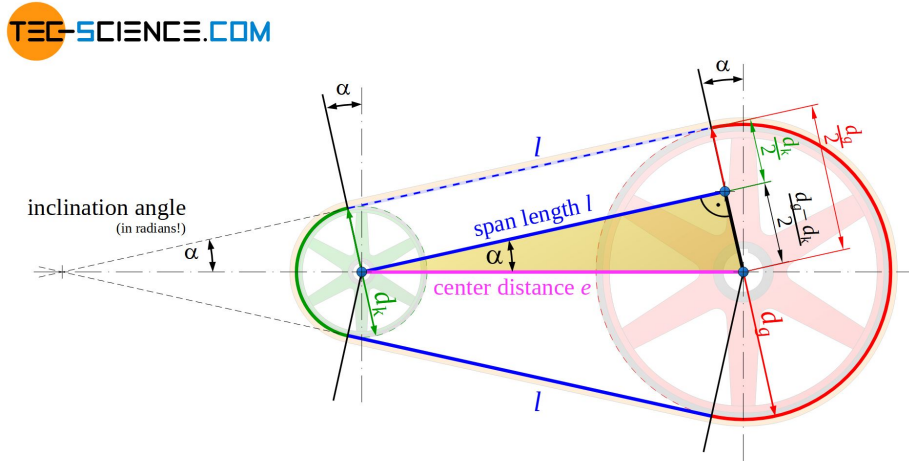


Figure 13: Belt geometry for two pulleys of radii r_p and r_s separated by center distance C . The belt wraps the pulleys with angles ϕ_p and ϕ_s and connects via two straight tangent spans ℓ .

Belt Length Decomposition The total belt length L_b is the sum of: (i) wrapped arc length on the primary pulley, (ii) wrapped arc length on the secondary pulley, and (iii) two straight tangent spans. Therefore,

$$L_b = r_p\phi_p + r_s\phi_s + 2\ell, \quad (7.2)$$

where ℓ is the length of a single straight tangent span segment.

Straight Span Length From the right-triangle geometry formed by the pulley centers and tangent points,

$$\ell = \sqrt{C^2 - (r_s - r_p)^2}. \quad (7.3)$$

Wrap Angles The wrap angles on the primary and secondary pulleys are

$$\phi_p = \pi - 2\lambda, \quad (7.4)$$

$$\phi_s = \pi + 2\lambda. \quad (7.5)$$

If $\lambda < 0$, these expressions automatically redistribute wrap between pulleys while preserving total belt length.

Tangency Angle (Signed Convention) Let λ denote the signed angle between the line of centers and the straight tangent belt segment (as defined in [Figure 13](#)). We define

$$\sin \lambda = \frac{r_s - r_p}{C}, \quad (7.6)$$

and equivalently,

$$\lambda = \sin^{-1}\left(\frac{r_s - r_p}{C}\right). \quad (7.7)$$

Under this convention, λ may be negative when $r_p > r_s$. This signed definition ensures the wrap distribution automatically adjusts when the pulley ordering reverses.

Closed Form Belt-Length Expression Substituting [Equation \(7.3\)](#) and [Equation \(7.4\)](#)–[Equation \(7.5\)](#) into [Equation \(7.2\)](#) yields

$$\begin{aligned} L_b &= r_p(\pi - 2\lambda) + r_s(\pi + 2\lambda) + 2\sqrt{C^2 - (r_s - r_p)^2} \\ &= \pi(r_p + r_s) + 2\lambda(r_s - r_p) + 2\sqrt{C^2 - (r_s - r_p)^2}. \end{aligned} \quad (7.8)$$

Finally, substituting [Equation \(7.7\)](#) gives

$$L_b = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2}. \quad (7.9)$$

Root Form (Geometric Compatibility Constraint) Rearranging Equation (7.9) yields

Belt Length Constraint Root Function (7.10)

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b.$$

The belt-length constraint is therefore

$$f(r_p, r_s, C) = 0. \quad (7.11)$$

Remark (Transcendental Nature and Approximation)

Equation (7.11) is transcendental due to the inverse-sine and square-root terms and therefore does not admit a closed-form solution.

In this work, the constraint is solved numerically using a bracketed root-finding method (e.g., bisection or Brent's method), which is deterministic and robust over the physically admissible interval.

It is common in simplified treatments to assume

$$\frac{r_s - r_p}{C} \approx 0,$$

which implies $\lambda \approx 0$ and therefore

$$\sin^{-1}\left(\frac{r_s - r_p}{C}\right) \approx 0.$$

Under this small-angle assumption, the belt-length equation reduces to a much simpler approximate form.

However, this approximation can introduce substantial geometric error when $\frac{r_s - r_p}{C}$ is not small. Appendix B quantifies this deviation and demonstrates that the approximation is not acceptable under realistic operating conditions.

7.5 Center-to-Center Distance Determination

In-order to make use of the belt-length constraint derived in Section 7.4, the center-to-center distance C must be determined.

The center-to-center distance C is a fixed geometric parameter of the assembled CVT system. Once the transmission is constructed, C does not vary during operation.

To determine C , we evaluate the belt-length constraint Equation (7.11) at the *known* initial configuration of the system.

Initial Geometric Configuration At startup, the CVT is assumed to be in its lowest ratio configuration:

- The primary pulley is at its minimum outer radius,

$$r_p = r_{p,\min}.$$

- The secondary pulley is at its maximum outer radius,

$$r_s = r_{s,\max}.$$

These values are determined by the physical geometry of the sheaves and represent the fully separated primary and fully closed secondary configuration.

Center Distance from Belt-Length Compatibility Substituting these known radii into Equation (7.11) yields

$$f(r_{p,\min}, r_{s,\max}, C) = 0. \quad (7.12)$$

This equation contains a single unknown, C , and is solved numerically to determine the fixed center-to-center distance consistent with the specified belt length L_b .

$$C = \text{root}_C f(r_{p,\min}, r_{s,\max}, C). \quad (7.13)$$

Remark (One-Time Geometric Solve)

The center distance C is determined once during model initialization. Thereafter, C is treated as a constant in all subsequent geometric and dynamic calculations.

7.6 Secondary Outer Radius as a Function of Shift

With the center-to-center distance C determined from Section 7.5, and the primary radius $r_p(s)$ defined by Equation (7.1), the secondary outer radius is obtained by enforcing the belt-length compatibility constraint.

Geometric Coupling At any shift position s , the primary radius $r_p(s)$ is known. The secondary radius r_s must therefore satisfy

$$f(r_p(s), r_s, C) = 0, \quad (7.14)$$

where $f(\cdot)$ is defined in Equation (7.10).

Numerical Determination of $r_s(s)$ Because Equation (7.14) is transcendental in r_s , the secondary radius is obtained numerically for each value of s :

Secondary Outer Radius

(7.15)

$$r_{s,\text{outer}}(s) = \text{root}_{r_s} f(r_p(s), r_s, C).$$

Remark (Dynamic Root Solve)

Unlike the center-distance calculation in [Section 7.5](#), which is performed once during initialization, this root solve is performed repeatedly as the shift coordinate s evolves during simulation. Because the physically admissible interval for r_s is bounded, a bracketed solver (e.g., Brent's method) converges rapidly.

With both $r_{p,\text{outer}}(s)$ and $r_{s,\text{outer}}(s)$ determined, all geometric quantities describing the belt path can now be expressed explicitly as functions of the shift coordinate s .

7.7 Wrap Angles as Functions of Shift

The wrap angles on the primary and secondary pulleys vary as the pulley radii change. Using the geometric relations from [Section 7.4](#), the tangency angle becomes

$$\lambda(s) = \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right). \quad (7.16)$$

Substitution into the wrap-angle definitions yields

$$\phi_p(s) = \pi - 2 \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right), \quad (7.17)$$

$$\phi_s(s) = \pi + 2 \sin^{-1} \left(\frac{r_{s,\text{outer}}(s) - r_{p,\text{outer}}(s)}{C} \right). \quad (7.18)$$

7.8 Effective Torque Radius

All geometric relations derived thus far use the outer surface of the belt as the reference radius. This simplifies the belt-length formulation and pulley geometry.

Torque transmission, however, occurs through the belt cross-section. The appropriate moment arm for torque is therefore located approximately at the mid-height of the belt, as illustrated in [Figure 14](#).

The effective torque radius for each pulley is defined as

$$r_{p,\text{eff}} = r_{p,\text{outer}} - \frac{h_b}{2}, \quad (7.19)$$

$$r_{s,\text{eff}} = r_{s,\text{outer}} - \frac{h_b}{2}. \quad (7.20)$$

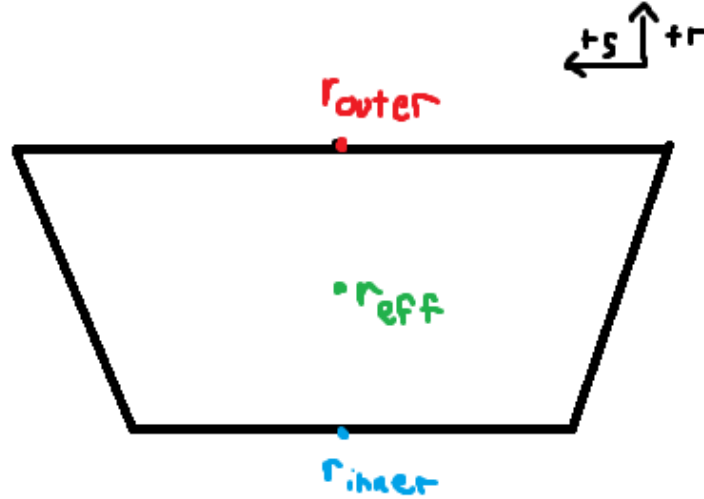


Figure 14: Belt cross-section showing the outer radius r_{outer} , inner radius r_{inner} , and effective torque radius r_{eff} located at the belt mid-height. The effective radius is used for all torque and power transmission calculations.

7.9 CVT Ratio Definition

The instantaneous transmission ratio is determined purely by pulley geometry. It is defined as the ratio of secondary to primary effective radii:

CVT Ratio	(7.21)
$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}.$	

Because both effective radii are functions of the shift coordinate s , the transmission ratio is likewise a function of s alone.

As the primary sheaves close, $r_{p,\text{eff}}(s)$ increases. By belt-length compatibility, the secondary radius decreases. Consequently, the ratio $R(s)$ decreases with increasing s .

Thus:

- Small s (minimum primary radius) corresponds to high ratio.
- Large s (maximum primary radius) corresponds to low ratio.

The CVT ratio is therefore a purely geometric quantity determined by the axial shift position.

7.10 Ratio Rate of Change

The CVT ratio is not only a function of shift position, but also changes in time whenever the sheaves are moving. This time variation matters because a changing ratio introduces additional kinematic coupling between the primary and secondary rotations. In other words, it is not sufficient to know the instantaneous ratio $R(s)$; the model also requires how quickly the ratio is changing.

This section derives an explicit expression for $\dot{R}$ in terms of the shift velocity $\dot{s}$ and geometric derivatives.

7.10.1 Start from the Ratio Definition

The ratio is defined by the effective radii as

$$R(s) = \frac{r_{s,\text{eff}}(s)}{r_{p,\text{eff}}(s)}. \quad (7.22)$$

Because R depends on time only through $s(t)$, the chain rule gives

$$\dot{R} = \frac{dR}{ds} \dot{s}. \quad (7.23)$$

Thus, the objective is to compute dR/ds .

7.10.2 Differentiate R with Respect to s

Differentiating Equation (7.22) with respect to s (quotient rule) gives

$$\frac{dR}{ds} = \frac{d}{ds} \left(\frac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \right) = \frac{\left(\frac{dr_{s,\text{eff}}}{ds} \right) r_{p,\text{eff}} - r_{s,\text{eff}} \left(\frac{dr_{p,\text{eff}}}{ds} \right)}{r_{p,\text{eff}}^2}. \quad (7.24)$$

Rearranging into a form that exposes the required components:

$$\frac{dR}{ds} = \frac{1}{r_{p,\text{eff}}} \frac{dr_{s,\text{eff}}}{ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \quad (7.25)$$

At this point, the remaining task is to express the two derivatives $dr_{p,\text{eff}}/ds$ and $dr_{s,\text{eff}}/ds$ in computable form.

7.10.3 Eliminating dr_s/ds Using the Belt-Length Constraint

The secondary radius $r_s(s)$ is defined implicitly through the belt-length constraint and does not admit a closed-form expression in terms of r_p . Direct differentiation of $r_s(s)$ is therefore not practical.

Instead, we differentiate the belt-length compatibility constraint derived in Section 7.4, which is written as

$$f(r_p, r_s, C) = 0, \quad (7.26)$$

with $f(\cdot)$ defined explicitly in Equation (7.10) and C constant. Since $r_p = r_p(s)$ and $r_s = r_s(s)$, differentiating Equation (7.26) with respect to s gives (multivariable chain rule)

$$\frac{\partial f}{\partial r_p} \frac{dr_p}{ds} + \frac{\partial f}{\partial r_s} \frac{dr_s}{ds} = 0. \quad (7.27)$$

Solving Equation (7.27) for dr_s/ds yields

$$\frac{dr_s}{ds} = -\frac{\frac{\partial f}{\partial r_p} dr_p}{\frac{\partial f}{\partial r_s} ds}. \quad (7.28)$$

7.10.4 Single Expression for dR/ds in Terms of Remaining Unknowns

Substituting Equation (7.28) into Equation (7.25) eliminates dr_s/ds and gives a single expression for dR/ds in terms of the belt constraint partial derivatives and the primary geometric derivative:

$$\frac{dR}{ds} = -\frac{1}{r_{p,\text{eff}}} \frac{\frac{\partial f}{\partial r_p} dr_p}{\frac{\partial f}{\partial r_s} ds} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \frac{dr_{p,\text{eff}}}{ds}. \quad (7.29)$$

Equation (7.29) makes the remaining requirements explicit. To evaluate dR/ds (and therefore $\dot{R}$ via Equation (7.23)), three quantities are required:

1. $\frac{\partial f}{\partial r_p}$,
2. $\frac{\partial f}{\partial r_s}$,
3. $\frac{dr_p}{ds}$ (and the associated $\frac{dr_{p,\text{eff}}}{ds}$).

The following subsections derive these three quantities for the specific belt-length function f defined in Equation (7.10) and for the primary geometry defined in Equation (7.1).

7.10.5 Partial Derivatives of the Belt Constraint

Recall the belt-length constraint written in root form as

$$f(r_p, r_s, C) = \pi(r_p + r_s) + 2(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + 2\sqrt{C^2 - (r_s - r_p)^2} - L_b = 0. \quad (7.30)$$

We now compute the two partial derivatives required in Equation (7.28), differentiating term-by-term and applying the product rule directly to the inverse-sine term.

Product rule for the inverse-sine factor Consider the factor

$$(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right).$$

Differentiating with respect to r_p (product rule) gives

$$\frac{\partial}{\partial r_p} \left[(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \right] = \underbrace{\frac{\partial}{\partial r_p} (r_s - r_p)}_{(1)} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p) \underbrace{\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right)}_{(2)}. \quad (7.31)$$

We therefore compute the two sub-derivatives.

Derivative (1)

$$\frac{\partial}{\partial r_p}(r_s - r_p) = -1. \quad (7.32)$$

Derivative (2) Using the chain rule,

$$\frac{d}{dx} \sin^{-1}(x) = \frac{1}{\sqrt{1-x^2}},$$

let

$$u = \frac{r_s - r_p}{C}.$$

Then

$$\frac{\partial u}{\partial r_p} = -\frac{1}{C}.$$

Therefore,

$$\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) = \frac{1}{\sqrt{1-u^2}} \left(-\frac{1}{C}\right). \quad (7.33)$$

Now simplify the denominator:

$$\sqrt{1-u^2} = \sqrt{1 - \left(\frac{r_s - r_p}{C}\right)^2} = \frac{\sqrt{C^2 - (r_s - r_p)^2}}{C}.$$

Substituting into [Equation \(7.33\)](#) yields

$$\frac{\partial}{\partial r_p} \sin^{-1}\left(\frac{r_s - r_p}{C}\right) = -\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}}. \quad (7.34)$$

Substitute into the product rule Substituting [Equation \(7.32\)](#) and [Equation \(7.34\)](#) into [Equation \(7.31\)](#) gives

$$\begin{aligned} \frac{\partial}{\partial r_p} \left[(r_s - r_p) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) \right] &= (-1) \sin^{-1}\left(\frac{r_s - r_p}{C}\right) + (r_s - r_p) \left(-\frac{1}{\sqrt{C^2 - (r_s - r_p)^2}} \right) \\ &= -\sin^{-1}\left(\frac{r_s - r_p}{C}\right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}}. \end{aligned} \quad (7.35)$$

Derivative of the square-root term For the straight-span contribution,

$$2\sqrt{C^2 - (r_s - r_p)^2},$$

we apply the chain rule:

$$\frac{\partial}{\partial r_p} \left[2\sqrt{C^2 - (r_s - r_p)^2} \right] = \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}. \quad (7.36)$$

Assemble $\partial f/\partial r_p$ Differentiating Equation (7.30) term-by-term gives

$$\begin{aligned}\frac{\partial f}{\partial r_p} &= \pi + 2 \left(-\sin^{-1} \left(\frac{r_s - r_p}{C} \right) - \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}} \right) + \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right).\end{aligned}\tag{7.37}$$

The square-root contributions cancel exactly.

Derivative with respect to r_s The same procedure yields

$$\frac{\partial}{\partial r_s} \left[(r_s - r_p) \sin^{-1} \left(\frac{r_s - r_p}{C} \right) \right] = \sin^{-1} \left(\frac{r_s - r_p}{C} \right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}},\tag{7.38}$$

and

$$\frac{\partial}{\partial r_s} \left[2\sqrt{C^2 - (r_s - r_p)^2} \right] = -\frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}}.\tag{7.39}$$

Assembling terms gives

$$\begin{aligned}\frac{\partial f}{\partial r_s} &= \pi + 2 \left(\sin^{-1} \left(\frac{r_s - r_p}{C} \right) + \frac{r_s - r_p}{\sqrt{C^2 - (r_s - r_p)^2}} \right) - \frac{2(r_s - r_p)}{\sqrt{C^2 - (r_s - r_p)^2}} \\ &= \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right).\end{aligned}\tag{7.40}$$

Summary of Belt Constraint Partial Derivatives From the derivations above, the partial derivatives of the belt-length constraint Equation (7.30) are

$$\frac{\partial f}{\partial r_p} = \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right),\tag{7.41}$$

$$\frac{\partial f}{\partial r_s} = \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right).\tag{7.42}$$

These expressions are globally valid under the geometric definition of λ given in Equation (7.7) and require no piecewise case distinction.

7.10.6 Primary Radius Derivative

From the piecewise primary radius model [Equation \(7.1\)](#), the primary outer radius satisfies

$$r_{p,\text{outer}}(s) = \begin{cases} r_{p,\text{min}}, & 0 \leq s < s_{\text{dz}}, \\ r_{p,\text{min}} + \frac{s - s_{\text{dz}}}{2 \tan \beta}, & s_{\text{dz}} \leq s \leq s_{\text{max}}. \end{cases}$$

Differentiating with respect to s gives

$$\frac{dr_p}{ds} = \frac{dr_{p,\text{outer}}}{ds} = \begin{cases} 0, & 0 \leq s < s_{\text{dz}}, \\ \frac{1}{2 \tan \beta}, & s_{\text{dz}} \leq s \leq s_{\text{max}}. \end{cases} \quad (7.43)$$

Because $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ and h_b is constant, the same derivative applies:

$$\frac{dr_{p,\text{eff}}}{ds} = \frac{dr_p}{ds}. \quad (7.44)$$

Remark (At the Travel Limit)

If the model clamps s at the physical limit s_{max} , then $r_{p,\text{outer}}$ (and thus $r_{p,\text{eff}}$) is constant beyond this point and the derivative is zero for $s > s_{\text{max}}$. In this document we restrict to the admissible domain $0 \leq s \leq s_{\text{max}}$.

7.10.7 Final Expression for dR/ds and $\dot{R}$

[Equation \(7.29\)](#) expresses dR/ds in terms of three quantities: (i) the belt-constraint partial derivatives $\partial f/\partial r_p$ and $\partial f/\partial r_s$, and (ii) the primary geometric derivatives dr_p/ds and $dr_{p,\text{eff}}/ds$.

From [Section 7.10.5](#), the partial derivatives of the belt-length constraint [Equation \(7.10\)](#) are

$$\frac{\partial f}{\partial r_p} = \pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right), \quad (7.45)$$

$$\frac{\partial f}{\partial r_s} = \pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right). \quad (7.46)$$

From [Section 7.10.6](#), the primary outer-radius derivative is [Equation \(7.43\)](#), and since $r_{p,\text{eff}} = r_{p,\text{outer}} - h_b/2$ with h_b constant, [Equation \(7.44\)](#) gives $dr_{p,\text{eff}}/ds = dr_p/ds$.

Substituting [Equation \(7.45\)](#)–[Equation \(7.46\)](#) and [Equation \(7.44\)](#) into [Equation \(7.29\)](#) yields

$$\frac{dR}{ds} = \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right] \frac{dr_p}{ds}. \quad (7.47)$$

Finally, substituting the piecewise primary derivative Equation (7.43) gives the fully expanded result

$$\frac{dR}{ds} = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ \frac{1}{2 \tan \beta} \left[-\frac{1}{r_{p,\text{eff}}} \frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} - \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}^2} \right], & s_{dz} \leq s \leq s_{\text{max}}. \end{cases} \quad (7.48)$$

Using the chain rule Equation (7.23), $\dot{R} = (dR/ds)\dot{s}$. Substituting Equation (7.48) and rearranging gives

Rate of Change of Transmission Ratio	(7.49)
$\dot{R} = \begin{cases} 0, & 0 \leq s < s_{dz}, \\ -\frac{\dot{s}}{2 \tan \beta r_{p,\text{eff}}} \left[\frac{\pi - 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)}{\pi + 2 \sin^{-1} \left(\frac{r_s - r_p}{C} \right)} + \frac{r_{s,\text{eff}}}{r_{p,\text{eff}}} \right], & s_{dz} \leq s \leq s_{\text{max}}. \end{cases}$	

Summary Equations Equation (7.48) and Equation (7.49) give the geometric sensitivity of the ratio to axial motion and its time rate of change. The deadzone behavior is explicit: when $0 \leq s < s_{dz}$, the ratio is constant and $\dot{R} = 0$. In the active region, $s_{dz} \leq s \leq s_{\text{max}}$, the ratio evolves according to geometry and the shift rate $\dot{s}$.

7.11 Section Summary and Forward Connection

This section established the complete *geometric foundation* of the CVT model. The development was intentionally kinematic: the objective was to derive a closed, computable mapping from axial shift to pulley geometry and transmission ratio, without introducing force balance, slip behavior, or torque relations.

At a high level, the geometry defines the chain

$$s(t) \longrightarrow r_{p,\text{eff}}(s), r_{s,\text{eff}}(s) \longrightarrow R(s) \longrightarrow \dot{R}(s, \dot{s}),$$

with each intermediate relationship derived explicitly from pulley and belt geometry.

Key Results Established The following geometric components are now defined and ready for use in subsequent modeling:

1. **Primary radius law.** Equation (7.1) provides the piecewise relationship between axial shift s and primary outer radius, including deadzone behavior and cone-angle dependence.
2. **Belt-length compatibility constraint.** Equation (7.11) enforces constant belt length and defines the implicit, nonlinear coupling between r_p , r_s , and the center distance C .

3. **Center distance determination.** Section 7.5 fixes the assembled center-to-center distance C by evaluating the belt constraint at a known geometric configuration. This solve is performed once during initialization.
4. **Secondary radius as a function of shift.** Section 7.6 defines $r_s(s)$ implicitly by solving the belt constraint at each shift position, producing the required redistribution of belt radius between pulleys.
5. **Wrap angles as functions of shift.** Section 7.7 defines the wrap angles $\phi_p(s)$ and $\phi_s(s)$ follow directly from the tangency geometry and are available for later contact, work, and belt-distribution calculations.
6. **Effective torque radii.** Section 7.8 corrects outer radii to effective radii using belt thickness, ensuring subsequent torque relations use an appropriate moment arm.
7. **Ratio and ratio rate.** Equation (7.21) defines the geometric ratio $R(s)$, and Section 7.10 provides $\frac{dR}{ds}$ and $\dot{R}$ via Equation (7.48) and Equation (7.49), including the deadzone condition where R is constant.

Structural Role in the Full Model These results form the geometric backbone of the CVT simulation. All subsequent sections treat the relationships derived here as fixed mappings. In particular, later torque and force models will use $r_{p,\text{eff}}(s)$ and $r_{s,\text{eff}}(s)$ as inputs, and shifting kinematics will use $R(s)$ and $\dot{R}(s, \dot{s})$ to relate primary and secondary motion during ratio change.

Several key properties are now clear: (i) the geometric coupling is inherently nonlinear (through the belt-length constraint), (ii) the deadzone is explicitly represented, and (iii) within the admissible travel range the mappings are differentiable and suitable for continuous-time simulation.

Transition to Dynamics With the geometry fully specified, the next objective is to determine how the shift coordinate evolves in time. Subsequent sections introduce axial force generation (from pulley mechanisms and belt interaction) to determine $\dot{s}$ and $\ddot{s}$. Once $\dot{s}$ is known, the ratio-rate expression Equation (7.49) provides the corresponding ratio evolution and the required kinematic coupling between rotating components during shifting.

8 Pulley Axial Force Models

This section introduces the axial force models that drive sheave motion and therefore determine the evolution of the shift coordinate $s(t)$.

The state vector $\mathbf{x}(t)$ was defined in Section 6 (see Equation (6.1)) as $(\omega_p, \omega_s, s, \dot{s})$. The objective of the present section is to derive expressions for the axial forces acting on the primary and secondary sheaves, which will ultimately provide the equation of motion for $\ddot{s}$.

Throughout this section, axial and radial directions follow the conventions established in Section 3.1: positive axial displacement corresponds to increasing primary effective radius, and radial forces are positive outward from the pulley center.

The goal is to express the net axial forces generated by the primary and secondary pulley mechanisms as functions of the current state. These forces will later be combined using Theory 1 (Newton's Second Law (Translation)) to obtain an equation of motion for s (and therefore $\dot{s}$ and $\ddot{s}$).

Although the specific models developed here correspond to the hardware considered in this work (flyweight/ramp actuation on the primary and a torque-reactive helix with springs on the secondary), the formulation is modular: alternative actuator models may be substituted provided they return the correct net axial force.

8.1 Ramp Geometry Parameterization

Both pulley actuators redirect forces through inclined surfaces. To keep the force models general and drop-in replaceable, we represent each mechanism's geometry using user-supplied functions of the axial shift coordinate s .

Primary Ramp Geometry (Axial-to-Radial Profile) On the primary pulley, centrifugal (radial) force is redirected into axial closing force through a ramp. We parameterize the ramp by an axial-to-radial profile

$$r_f = r_f(s), \quad (8.1)$$

where $r_f(s)$ is the effective flyweight radius.

Because s is axial distance, the local ramp slope is dr_f/ds . The corresponding ramp angle is defined consistently as

$$\alpha_p(s) \equiv \tan^{-1}\left(\frac{dr_f}{ds}\right), \quad \text{so that} \quad \tan \alpha_p(s) = \frac{dr_f}{ds}. \quad (8.2)$$

This definition is physically intuitive: a steeper ramp yields a larger change in radius per unit axial travel and therefore larger axial gain through the $\tan \alpha_p$ factor used later in the force model.

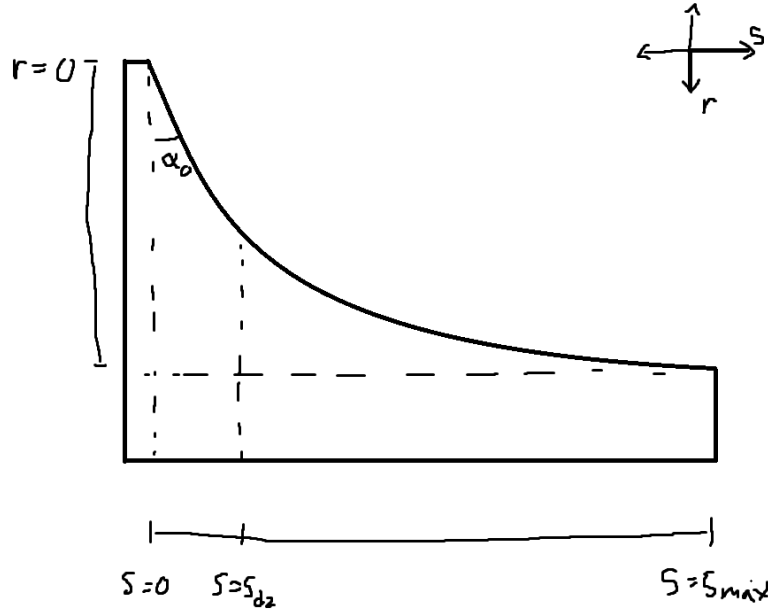


Figure 15: Primary ramp geometry. The flyweight radius is parameterized by $r_f(s)$, where s is axial travel. The local ramp angle is defined by $\tan \alpha_p(s) = dr_f/ds$, so steeper profiles (larger dr_f/ds) redirect a larger fraction of radial force into axial closing force.

Primary Admissibility Conditions The primary ramp profile $r_f(s)$ must satisfy:

1. $r_f(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\frac{dr_f}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\frac{dr_f}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the ramp angle $\alpha_p(s)$ is well-defined and continuous. Condition (2) guarantees that increasing axial displacement does not reduce flyweight radius. Condition (3) ensures the ramp angle remains strictly below $\pi/2$ and the axial gain $\tan \alpha_p$ remains finite.

Secondary Helix Geometry (Axial-to-Rotation Mapping) On the secondary pulley, axial travel produces a relative rotation of the helix mechanism, loading the torsional spring and generating torque-reactive clamping. We parameterize this kinematically as

$$\theta_s = \theta_s(s), \quad (8.3)$$

where $\theta_s(s)$ is the helix-induced rotation.

The quantity $d\theta_s/ds$ is the helix “lead” (rotation per unit axial travel). Under the convention used in this work, *shallower* helix ramps produce *more* rotation for the same axial travel, while *steeper* ramps produce *less* rotation. To encode this consistently, we define the effective helix angle $\alpha_s(s)$ by the inverse-slope relationship

$$\tan \alpha_s(s) \equiv \frac{1}{r_h \frac{d\theta_s}{ds}}, \quad (8.4)$$

where r_h is the effective helix radius. With this definition, a decrease in α_s (a “shallower” ramp) corresponds to an increase in $d\theta_s/ds$, matching the physical behavior described above.

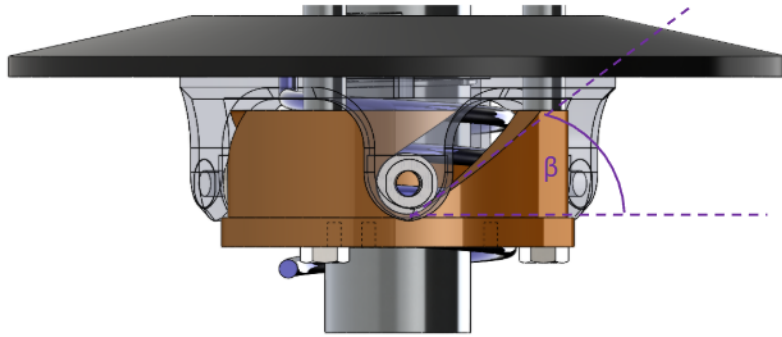


Figure 16: Secondary helix geometry. Axial travel s induces rotation $\theta_s(s)$ of the helix mechanism. The helix angle is defined so that shallower ramps produce greater rotation per unit axial travel: $\tan \alpha_s(s) = (r_h d\theta_s/ds)^{-1}$. Diagram should be 3d, show clearly which direction shift distance is in (and why the angle is defined differently). Use dotted lines to show a single angle along the ramp (instantaneous slope or smt)

Secondary Admissibility Conditions The helix rotation mapping $\theta_s(s)$ must satisfy:

1. $\theta_s(s) \in C^1([0, s_{\max}])$ (continuously differentiable),
2. $\frac{d\theta_s}{ds} \geq 0$ over $[0, s_{\max}]$,
3. $\frac{d\theta_s}{ds}$ is finite over $[0, s_{\max}]$.

Condition (1) ensures the torque-to-axial conversion remains continuous. Condition (2) guarantees that increasing axial displacement increases torsional loading. Condition (3) ensures the effective helix angle remains strictly below $\pi/2$ and the geometric gain remains finite.

Remark (Piecewise Profiles in Practice)

The C^1 requirement above is sufficient for all derivations in this document. In practice, piecewise C^1 profiles (continuous everywhere, continuously differentiable except at finitely many junction points where left and right derivatives exist) are also admissible for numerical simulation. At junction points, one-sided derivatives provide the required slope values, and the resulting force discontinuities pose no difficulty for standard ODE solvers. However, large slope discontinuities can introduce local stiffness into the system: the abrupt change in force gradients may require adaptive solvers to reduce the timestep significantly near the junction, degrading computational efficiency. In extreme cases, explicit integrators may struggle to maintain stability without impractically small step sizes, and implicit or stiff-aware solvers may be preferable.

Derivations for common ramp and helix profiles are provided in Appendix ??.

8.2 Primary Axial Force Model

The primary axial clamping force arises from centrifugal loading of the flyweights and opposition from the primary compression spring.

Figure 17: Primary CVT assembly showing flyweight mass m_f , ramp profile $r_f(s)$, compression spring with stiffness k_p and preload $x_{p,0}$, and axial shift coordinate s . TODO: use 3D model to define constants and show various aspects of primary CVT.

The state variable governing primary rotation is ω_p .

Centrifugal Loading of Flyweights From [Theory 5 \(Centrifugal Force\)](#),

$$F_c = m\omega^2 r.$$

Let

- m_f denote the flyweight mass,
- $r_f(s)$ denote the effective flyweight radius.

The radial centrifugal force is therefore

$$F_{p,\text{rad}}(s, \omega_p) = m_f \omega_p^2 r_f(s). \quad (8.5)$$

Radial-to-Axial Projection The ramp redirects radial force into axial force. As shown in Figure 18, the axial component of a force acting along the ramp is obtained by

$$F_{p,\text{axial component}} = F_{p,\text{rad}} \tan \alpha_p(s). \quad (8.6)$$

From Equation (8.2), the ramp angle satisfies

$$\tan \alpha_p(s) = \frac{dr_f}{ds}.$$

Substituting this into Equation (8.6) gives

$$F_{p,\text{cent}}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds}. \quad (8.7)$$

Thus the axial gain of centrifugal force is directly governed by the local slope of the ramp profile.

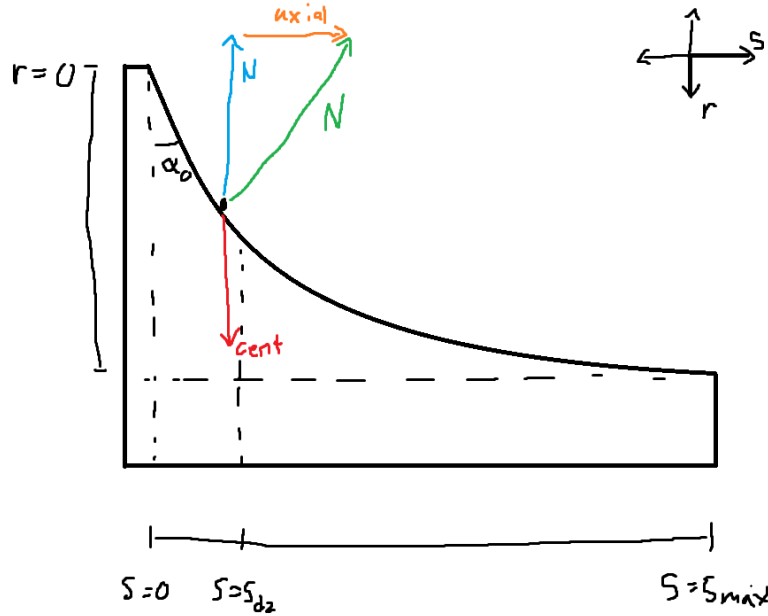


Figure 18: Primary ramp geometry. Radial centrifugal force is redirected into axial force through the ramp. The projection introduces a factor of $\tan \alpha_p$, which equals dr_f/ds by definition of the ramp slope.

Primary Compression Spring From **Theory 6 (Hooke's Law (Compressional))**,

$$F = k\Delta x.$$

Let k_p denote primary stiffness and $x_{p,0}$ preload.

$$F_{p,\text{spring}}(s) = k_p(x_{p,0} + s). \quad (8.8)$$

Net Primary Axial Force The net axial force generated by the primary mechanism is the difference between the centrifugal axial component and the spring force:

Primary Axial Force $F_{p,\text{ax}}(s, \omega_p) = m_f \omega_p^2 r_f(s) \frac{dr_f}{ds} - k_p(x_{p,0} + s).$	(8.9)
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8.3 Secondary Axial Force Model

The secondary pulley is *torque-reactive*: its axial clamping force is generated in response to torque transmitted through the CVT, rather than from rotational speed (as on the primary). This fundamental difference determines the load-sensitive behavior of the secondary mechanism.

Figure 19: Secondary CVT assembly showing the helix mechanism, torsional spring with stiffness $k_{s,\theta}$ and preload $\theta_{s,0}$, compression spring with stiffness $k_{s,x}$ and preload $x_{s,0}$, and effective helix radius r_h . The torque τ_s transmitted through the belt creates a tangential force at the helix, which is redirected into axial clamping force through the inclined helix ramps. TODO: use 3D model to show helix geometry, torque path, and direction of resulting axial force.

Why the Secondary Is Torque-Reactive On the primary pulley, centrifugal force from the spinning flyweights generates axial clamping—this depends on rotational *speed* (ω_p), not on the torque being transmitted. The primary therefore tends to upshift at high RPM regardless of load.

The secondary operates on a fundamentally different principle: it generates axial clamping force in response to torque *transmitted through the CVT*. To understand why, consider the mechanics of force generation at the helix.

The helix mechanism converts tangential force into axial force. However, a tangential force can only exist if there is resistance to rotation. If the secondary pulley were free to spin (i.e., no load on the driven shaft), the belt would simply rotate the pulley without creating any tangential force at the helix contact surface—and therefore no axial clamping would occur.

For a reaction force to exist at the helix, the following must hold:

- The belt must apply a driving torque to the secondary pulley (higher tension on the entry side than the exit side).
- The driven load must resist this torque, preventing free rotation.

- The torque transmitted from belt to load passes through the helix mechanism, creating a tangential force at the helix ramps.
- This tangential force, acting on the inclined helix surface, generates an axial component that clamps the sheaves together.

In other words, the helix “reacts” against the torque being transmitted. The harder the CVT must work (i.e., the more torque flows through the secondary to the load), the harder the secondary clamps the belt.

This distinction is critical:

- The relevant torque is the torque *transmitted through the CVT* to the driven load—not simply the load torque or an externally applied torque.
- If the driven shaft were disconnected, the secondary would spin freely and generate no clamping force, regardless of belt tension.
- Under light load, secondary clamping is weak, allowing the primary to dominate and upshift. Under heavy load, secondary clamping increases, resisting upshift or forcing a downshift.

This torque-reactive behavior provides automatic load sensitivity: the CVT naturally downshifts under high-torque conditions (e.g., hill climbing) and upshifts under light-torque conditions (e.g., cruising).

Definition of Secondary Shaft Torque Let τ_s denote the torque transmitted through the secondary shaft. This is the net torque that flows from the belt, through the secondary pulley mechanism (including the helix), and into the driven load. It is this transmitted torque—not simply the load seen by the shaft—that creates the reaction force at the helix and therefore determines the axial clamping force.

The relationship between τ_s and the belt tension distribution is developed in the following section on belt mechanics; for the present derivation, τ_s is treated as a known input to the secondary axial force model.

Torsional Spring Torque From [Theory 7 \(Hooke’s Law \(Torsional\)\)](#),

$$\tau = k\Delta\theta.$$

Let $k_{s,\theta}$ denote torsional stiffness and $\theta_{s,0}$ preload.

$$\tau_{s,\text{spring}}(s) = k_{s,\theta}(\theta_{s,0} + \theta_s(s)). \quad (8.10)$$

Total Torque Applied to Helix

$$\tau_{s,\text{total}} = \tau_s + \tau_{s,\text{spring}}(s). \quad (8.11)$$

Both torque contributions act in the same rotational direction at the helix: the transmitted torque τ_s loads the helix as it passes from belt to driven shaft, while the torsional spring $\tau_{s,\text{spring}}$ resists the relative rotation induced by axial displacement. Both therefore contribute positively to the tangential force at the helix ramps, and hence to axial clamping. The torsional spring provides

a baseline clamping force even when no torque is being transmitted ($\tau_s = 0$), ensuring the belt remains seated at idle.

Torque-to-Axial Conversion Through the Helix From [Theory 3 \(Torque Definition\)](#), $\tau = F_t r_h$, so

$$F_t = \frac{\tau_{s,\text{total}}}{r_h}.$$

The helix redirects tangential force into axial force. From trigonometry (see [Figure 20](#)),

$$F_{s,\text{helix}} = \frac{F_t}{2 \tan \alpha_s(s)}. \quad (8.12)$$

From [Section 8.1](#), the helix angle satisfies

$$\tan \alpha_s(s) = \frac{1}{r_h \frac{d\theta_s}{ds}}, \quad \Rightarrow \quad \frac{1}{\tan \alpha_s(s)} = r_h \frac{d\theta_s}{ds}.$$

Substituting into [Equation \(8.12\)](#) yields

$$F_{s,\text{helix}}(s, \tau_s) = \frac{\tau_{s,\text{total}}}{2} \frac{d\theta_s}{ds}. \quad (8.13)$$

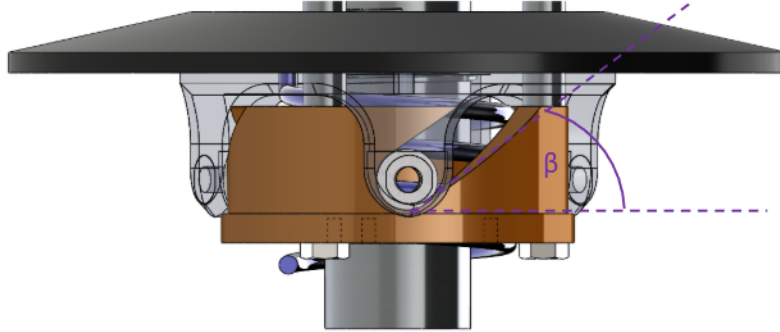


Figure 20: Secondary helix geometry. Tangential force generated by shaft torque is redirected into axial force. The projection introduces a factor $1/\tan \alpha_s$, which equals $r_h d\theta_s/ds$ by definition of the helix mapping. TODO: Add a point with forces (FBD)

Secondary Compression Spring In addition to helix loading, the secondary includes a compression spring. From [Theory 6 \(Hooke's Law \(Compressional\)\)](#),

$$F = k\Delta x.$$

Let $k_{s,x}$ denote the secondary compression stiffness and $x_{s,0}$ preload. With compression increasing with s ,

$$F_{s,\text{comp}}(s) = k_{s,x}(x_{s,0} + s). \quad (8.14)$$

Net Secondary Axial Force The net secondary axial force is the sum of the helix-generated clamping force and the compression spring force:

Secondary Axial Force $F_{s,\text{ax}}(s, \tau_s) = \underbrace{\frac{\tau_s + k_{s,\theta}(\theta_{s,0} + \theta_s(s))}{2} \frac{d\theta_s}{ds}}_{\text{helix + torsional spring}} + \underbrace{k_{s,x}(x_{s,0} + s)}_{\text{compression spring}}.$	(8.15)
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8.4 Belt Centrifugal Force on the Wrap

In addition to the actuator and spring forces derived previously, the belt itself generates centrifugal loading as it rotates around each pulley. This section derives the resulting contribution to axial clamping force.

Centrifugal Force Formulation From [Theory 5 \(Centrifugal Force\)](#), a mass m rotating at angular velocity ω at radius r experiences centrifugal force

$$F_c = m\omega^2 r.$$

To apply this to the belt wrapped around a pulley, we require:

1. The mass of belt material in contact with the pulley.
2. The radius at which this mass rotates.
3. The angular velocity of the belt.

Each of these quantities is derived below.

Belt Cross-Section Geometry The belt cross-section is approximated as a trapezoid (see [Figure 21](#)) with:

- radial thickness h_b ,
- outer width b_{out} (at the belt's outer surface),
- inner width b_{in} (at the belt's inner surface).

Figure 21: Trapezoidal belt cross-section. The outer width b_{out} corresponds to the surface away from the pulley axis; the inner width b_{in} corresponds to the surface closer to the axis. The radial thickness is h_b .

The cross-sectional area of the trapezoid is

$$A_b = \frac{h_b}{2} (b_{\text{out}} + b_{\text{in}}). \quad (8.16)$$

Let ρ_b denote the belt material density (kg/m³). The linear mass density (mass per unit belt length) is then

$$\mu_b = \rho_b A_b. \quad (8.17)$$

Belt Mass Centroid Centrifugal force acts at the center of mass of the rotating belt element, not at the outer or inner surface. Because the belt cross-section is trapezoidal (not rectangular), the centroid is offset from the geometric center.

Let y denote radial distance measured inward from the outer surface, so that $y = 0$ at the outer surface and $y = h_b$ at the inner surface. For a trapezoid with linear sidewalls, the local width varies linearly with y :

$$b(y) = b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y. \quad (8.18)$$

The centroid location Δr_{cm} , measured inward from the outer surface, is computed from the definition of the centroid of an area:

$$\Delta r_{\text{cm}} = \frac{1}{A_b} \int_0^{h_b} y b(y) \, dy. \quad (8.19)$$

Substituting [Equation \(8.18\)](#) into this integral:

$$\begin{aligned} \int_0^{h_b} y b(y) \, dy &= \int_0^{h_b} y \left[b_{\text{out}} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} y \right] \, dy \\ &= \int_0^{h_b} b_{\text{out}} y \, dy + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \int_0^{h_b} y^2 \, dy \\ &= b_{\text{out}} \left[\frac{y^2}{2} \right]_0^{h_b} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \left[\frac{y^3}{3} \right]_0^{h_b} \\ &= b_{\text{out}} \cdot \frac{h_b^2}{2} + \frac{b_{\text{in}} - b_{\text{out}}}{h_b} \cdot \frac{h_b^3}{3} \\ &= \frac{h_b^2}{2} b_{\text{out}} + \frac{h_b^2}{3} (b_{\text{in}} - b_{\text{out}}) \\ &= \frac{h_b^2}{6} (3b_{\text{out}} + 2b_{\text{in}} - 2b_{\text{out}}) \\ &= \frac{h_b^2}{6} (b_{\text{out}} + 2b_{\text{in}}). \end{aligned} \quad (8.20)$$

Dividing by the area $A_b = \frac{h_b}{2}(b_{\text{out}} + b_{\text{in}})$ from Equation (8.16):

$$\Delta r_{\text{cm}} = \frac{\frac{h_b^2}{6}(b_{\text{out}} + 2b_{\text{in}})}{\frac{h_b}{2}(b_{\text{out}} + b_{\text{in}})} = h_b \cdot \frac{b_{\text{out}} + 2b_{\text{in}}}{3(b_{\text{out}} + b_{\text{in}})}. \quad (8.21)$$

The belt mass-centroid radius at any shift position is therefore

$$r_{\text{cm}}(s) = r_{\text{out}}(s) - \Delta r_{\text{cm}}, \quad (8.22)$$

where $r_{\text{out}}(s)$ is the outer belt radius at the current shift (equal to the pulley outer radius at that position).

Distributed Centrifugal Load Along the Wrap Consider a differential belt element along the wrapped arc. Let θ denote the angular coordinate measured from the center of the wrap, so that the wrapped arc spans

$$\theta \in \left[-\frac{\phi}{2}, \frac{\phi}{2} \right],$$

where ϕ is the total wrap angle (see Figure 22).

Figure 22: Belt wrapped around a pulley. A differential element at angle θ has arc length $d\ell = r_{\text{cm}} d\theta$ and experiences radially outward centrifugal force.

A differential belt element has arc length

$$d\ell = r_{\text{cm}} d\theta,$$

and therefore differential mass

$$dm = \mu_b d\ell = \mu_b r_{\text{cm}} d\theta. \quad (8.23)$$

From Theory 5 (Centrifugal Force), the centrifugal force on this element is

$$dF_c = dm \omega^2 r_{\text{cm}} = \mu_b \omega^2 r_{\text{cm}}^2 d\theta. \quad (8.24)$$

This force acts radially outward at each point along the wrap.

Total Radial Force Over the Wrap Integrating Equation (8.24) over the entire wrap angle yields the total radially-directed centrifugal force:

$$\begin{aligned}
F_{c,\text{rad}} &= \int_{-\phi/2}^{\phi/2} \mu_b \omega^2 r_{\text{cm}}^2 \, d\theta \\
&= \mu_b \omega^2 r_{\text{cm}}^2 \int_{-\phi/2}^{\phi/2} d\theta \\
&= \mu_b \omega^2 r_{\text{cm}}^2 \phi \\
&= \rho_b A_b \omega^2 r_{\text{cm}}^2 \phi.
\end{aligned} \tag{8.25}$$

Radial-to-Axial Conversion Through the Sheave Cone The radial centrifugal force acts outward against the conical sheave faces. Because the sheaves are inclined at cone half-angle β , this radial loading is redirected into an axial component that tends to push the sheaves together (i.e., contributes to clamping force).

From Equation (7.43), the relationship between radial and axial displacement in the active shift region is

$$\frac{dr}{ds} = \frac{1}{2 \tan \beta}.$$

This geometric ratio determines how radial force projects onto the axial direction. Inverting, a radial force F_{rad} produces an axial clamping contribution

$$F_{\text{ax}} = F_{\text{rad}} \cdot \frac{1}{2 \tan \beta}.$$

Substituting the radial centrifugal force from Equation (8.25):

$$F_{c,\text{ax}} = \frac{\rho_b A_b \omega^2 r_{\text{cm}}^2 \phi}{2 \tan \beta}. \tag{8.26}$$

Primary Pulley Contribution On the primary pulley, the belt rotates at approximately $\omega = \omega_p$. The outer belt radius equals the primary outer radius from Equation (7.1), which is piecewise in s and accounts for the deadzone behavior.

The mass-centroid radius is

$$r_{p,\text{cm}}(s) = r_{p,\text{outer}}(s) - \Delta r_{\text{cm}}, \tag{8.27}$$

and the wrap angle $\phi_p(s)$ is given by ??.

Substituting into Equation (8.26) yields the primary belt centrifugal contribution to axial clamping:

$$F_{c,p}(s, \omega_p) = \frac{\rho_b A_b \omega_p^2 r_{p,\text{cm}}(s)^2 \phi_p(s)}{2 \tan \beta}. \quad (8.28)$$

Remark (Belt Angular Velocity)

Using the primary angular velocity ω_p in the centrifugal force calculation is an approximation that assumes the belt speed matches the primary sheave speed. In reality, belt slip and compliance may cause the belt to rotate at a vastly different angular velocity than the primary sheave.

Secondary Pulley Contribution On the secondary pulley, the belt rotates at approximately $\omega = \omega_s$. The outer belt radius $r_{s,\text{outer}}(s)$ is determined implicitly by the belt-length constraint (see [Equation \(7.15\)](#)).

The mass-centroid radius is

$$r_{s,\text{cm}}(s) = r_{s,\text{outer}}(s) - \Delta r_{\text{cm}}, \quad (8.29)$$

and the wrap angle $\phi_s(s)$ is given by ??.

The secondary belt centrifugal contribution to axial clamping is

$$F_{c,s}(s, \omega_s) = \frac{\rho_b A_b \omega_s^2 r_{s,\text{cm}}(s)^2 \phi_s(s)}{2 \tan \beta}. \quad (8.30)$$

8.5 Axial Shift Dynamics

We now assemble the axial force models into a single equation of motion for the shift coordinate $s(t)$, defined in [Section 6](#) as part of the state vector $(\omega_p, \omega_s, s, \dot{s})$. As established in [Section 7.11](#), the belt-length constraint and conical kinematics fully determine all pulley radii, wrap angles, and span geometry once s is specified, so s serves as a single generalized coordinate for the entire shifting system.

Sign Convention By definition, increasing s corresponds to closing the primary pulley. Because total belt length is fixed, this necessarily induces the complementary motion at the secondary. Under this convention:

- A positive primary axial force $F_{p,\text{ax}}$ tends to increase s .
- A positive secondary axial force $F_{s,\text{ax}}$ tends to decrease s .
- The belt centrifugal forces $F_{c,p}$ and $F_{c,s}$ act in the same direction as their respective pulley's axial force (i.e., $F_{c,p}$ adds to the primary's tendency to increase s , while $F_{c,s}$ adds to the secondary's tendency to decrease s).

This sign structure will be reflected explicitly in the final force balance.

Equivalent Inertia Associated with s Newton's second law applies to physical masses in translation. Since s parameterizes coordinated motion of several bodies, we construct an equivalent mass m_{eq} such that the total kinetic energy associated with shift can be written in the form of [Theory 4 \(Kinetic Energy \(Translational\)\)](#):

$$T_{\text{shift}} = \frac{1}{2} m_{\text{eq}} \dot{s}^2.$$

The movable primary sheave translates with speed $\dot{s}$ in the positive coordinate direction. The movable secondary sheave translates with equal magnitude speed (but opposite physical direction), so its kinetic energy also contributes proportionally to $\dot{s}^2$.

Let $m_{p,m}$ and $m_{s,m}$ denote the masses of the movable primary and secondary sheaves.

Belt Mass Contribution The total belt mass is

$$m_b = \rho_b A_b L_b, \quad (8.31)$$

where A_b is the trapezoidal cross-sectional area from [Equation \(8.16\)](#) and L_b is the constant belt length.

The belt moves continuously with a single belt speed, but the axial component of that motion associated with a change in s is determined by geometry. From the axial-radial coupling in [Equation \(7.43\)](#), a change in sheave separation s is shared between the two conical faces supporting the belt wedge. Thus, the belt's axial velocity relative to the pulley axis is

$$\dot{s}_b = \frac{\dot{s}}{2}. \quad (8.32)$$

The kinetic energy associated with this axial motion is therefore

$$T_b = \frac{1}{2} m_b \left(\frac{\dot{s}}{2} \right)^2 = \frac{1}{2} \left(\frac{m_b}{4} \right) \dot{s}^2.$$

Expressed in the generalized coordinate s , the belt contributes an effective mass of $m_b/4$.

Total Equivalent Mass The total inertia associated with shift becomes

$$m_{\text{eq}} = m_{p,m} + m_{s,m} + \frac{m_b}{4} = m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}. \quad (8.33)$$

Net Axial Force in the s Coordinate Collecting the axial force contributions derived previously:

- $F_{p,\text{ax}}(s, \omega_p)$ from [Equation \(8.9\)](#),
- $F_{s,\text{ax}}(s, \tau_s)$ from [Equation \(8.15\)](#),
- $F_{c,p}(s, \omega_p)$ from [Equation \(8.28\)](#),
- $F_{c,s}(s, \omega_s)$ from [Equation \(8.30\)](#).

With the sign convention defined above, the generalized force in the positive s direction is

$$\sum F_s = (F_{p,ax} + F_{c,p}) - (F_{s,ax} + F_{c,s}). \quad (8.34)$$

Shift Equation of Motion Applying Newton's second law in the generalized coordinate,

$$\sum F_s = m_{eq} \ddot{s},$$

and substituting Equation (8.33) and Equation (8.34), we obtain

Axial Shift Acceleration (8.35) $\ddot{s} = \frac{(F_{p,ax}(s, \omega_p) + F_{c,p}(s, \omega_p)) - (F_{s,ax}(s, \tau_s) + F_{c,s}(s, \omega_s))}{m_{p,m} + m_{s,m} + \frac{\rho_b A_b L_b}{4}}.$
--

Equation (8.35) provides the axial shift acceleration as a function of the current state variables $(s, \dot{s}, \omega_p, \omega_s)$ and the reacted secondary torque τ_s . The remaining coupling between axial and rotational dynamics enters through τ_s , which will be determined later using Theory 2 (Newton's Second Law (Rotation)).

8.6 Section Summary and Forward Outlook

In this section we established a complete axial force model for the CVT mechanism and derived a closed-form equation of motion for the generalized shift coordinate $s(t)$.

Beginning with geometric parameterizations of the primary ramp and secondary helix, we expressed all actuator forces in terms of the state variables $(s, \dot{s}, \omega_p, \omega_s)$. We then incorporated spring forces, torque-reactive loading on the secondary, and the distributed centrifugal loading of the belt. Finally, by constructing an equivalent inertia associated with the coordinated axial motion of the movable sheaves and belt mass, we obtained the governing shift dynamics equation Equation (8.35).

At this stage, the axial subsystem is formally closed *except* for one remaining coupling quantity: the reacted secondary torque τ_s . Although τ_s appears explicitly in the secondary axial force model, its value is determined by rotational dynamics of the drivetrain and by torque transmission through the belt. Aside from the shared state variables, the axial and rotational subsystems are coupled only through this torque term.

The next step is to determine how torque is transmitted across the CVT as a function of pulley radii and speed ratio. This will establish the relationship between τ_s , input torque at the primary, and the evolving ratio $R = r_s/r_p$. Once this transmission relationship is obtained, the axial and rotational equations can be combined into a unified, state-space representation of the full CVT dynamics.

With the axial force structure now complete, attention turns to the torque transmission mechanism that closes the remaining dynamical loop.

9 Torque Transmission and Coupling Torque

The axial shift dynamics derived in [Section 8.5](#) require the reacted secondary torque τ_s appearing in [Equation \(8.15\)](#). In this section we determine the torque transmitted through the CVT as a function of the rotational states and the instantaneous ratio $R(t)$.

We proceed in four steps:

1. Write rotational equations of motion for the primary and secondary pulleys.
2. Define the inertias associated with each pulley.
3. Introduce the CVT ratio kinematics.
4. Eliminate accelerations to obtain the transmitted torque.

The result provides the ideal torque transmission law associated with the geometric ratio $R(t)$. This serves as the baseline torque model and will be generalized in the next section.

9.1 System Definition and Torque Path

Figure 23: Torque transmission schematic (placeholder). Engine torque τ_{eng} drives the primary pulley (angular velocity ω_p). Coupling torque τ_p is transmitted through the belt to the secondary pulley (angular velocity ω_s), where it appears as $\tau_s = R\tau_p$. The secondary delivers torque to the downstream drivetrain, which resists motion with torque τ_{load} .

We model two rotating bodies:

- Primary pulley (rigidly connected to the engine),
- Secondary pulley (driving the downstream drivetrain and vehicle).

Let

- I_p : total rotational inertia of the engine + primary CVT pulley,
- I_s : total rotational inertia about the secondary pulley,
- $\tau_{\text{eng}}(t)$: engine output torque applied to the primary pulley, modeled as a user-supplied function of primary speed (see [Section 9.3](#)),
- $\tau_p(t)$: coupling torque (as seen at the primary pulley),
- $\tau_s(t)$: coupling torque (as seen at the secondary pulley),
- $\tau_{\text{load}}(t)$: resisting torque reflected to the secondary pulley from road loads (see [Section 9.4](#)),

9.2 Rotational Equations of Motion

Applying [Theory 2 \(Newton's Second Law \(Rotation\)\)](#) to each pulley:

Primary Pulley

$$\tau_{\text{eng}}(t) - \tau_p(t) = I_p \alpha_p(t), \quad (9.1)$$

Secondary Pulley

$$\tau_s(t) - \tau_{\text{load}}(t) = I_s \alpha_s(t), \quad (9.2)$$

where

$$\alpha_p(t) = \frac{d\omega_p}{dt}, \quad \alpha_s(t) = \frac{d\omega_s}{dt}.$$

At this stage we have four unknown quantities:

$$\alpha_p(t), \quad \alpha_s(t), \quad \tau_p(t), \quad \tau_s(t),$$

but only two equations (Equation (9.1) and Equation (9.2)). Two additional relations are therefore required to close the system. First, we will define the inertias I_p and I_s in terms of physical parameters of the system. At this point, the external torques $\tau_{\text{eng}}(t)$ and $\tau_{\text{load}}(t)$ remain to be specified. In the remainder of this section we: (i) define τ_{eng} from the engine torque curve under a full-throttle assumption, and (ii) derive τ_{load} from longitudinal road-load theories and reflect it to the secondary pulley.

9.3 Engine Torque Model

The primary forcing term $\tau_{\text{eng}}(t)$ is supplied by the engine. In this formulation, the engine is modeled as a user-provided torque curve defined as a function of primary angular velocity.

Following Assumption 14 (Full-Throttle Engine Model), we assume full-throttle operation throughout the simulation. Under this assumption,

$$\tau_{\text{eng}}(t) = \tau_{\text{eng}}(\omega_p(t)), \quad (9.3)$$

where $\tau_{\text{eng}}(\cdot)$ is a supplied admissible function.

Admissibility Constraints. The engine torque function must satisfy:

- It is defined over a bounded operating domain $\omega_{\min} \leq \omega_p \leq \omega_{\max}$.
- It is finite and real-valued on this domain.
- Outside this domain, torque is either saturated or ω_p is constrained to remain within bounds.

No additional structural assumptions (e.g., polynomial form, closed-form analytic expression) are required by the dynamical model. Only the mapping $\omega_p \mapsto \tau_{\text{eng}}$ is needed to evaluate Equation (9.1).

9.4 Load Torque Model

The resisting torque $\tau_{\text{load}}(t)$ in Equation (9.2) represents the longitudinal road load reflected to the secondary pulley.

We derive this torque from first principles: (i) force balance on an inclined vehicle, (ii) standard rolling resistance theory, (iii) quadratic aerodynamic drag, (iv) reflection through wheel radius and gear ratio.

Step 1: Wheel and Gear Reflection Let r_w denote effective wheel rolling radius, and $G > 0$ the fixed gear ratio defined so that

$$\omega_s = G \omega_w.$$

Vehicle speed is therefore

$$v(t) = r_w \omega_w(t) = \frac{r_w}{G} \omega_s(t). \quad (9.4)$$

If $F_{\text{road}}(t)$ denotes the total longitudinal resistive force on the vehicle, then the wheel torque is

$$\tau_{\text{road},w}(t) = r_w F_{\text{road}}(t). \quad (9.5)$$

Assuming ideal power transmission (**Assumption 12 (Ideal CVT Power Transmission)**), the reflected secondary torque is

$$\tau_{\text{load}}(t) = \frac{1}{G} \tau_{\text{road},w}(t) = \frac{r_w}{G} F_{\text{road}}(t). \quad (9.6)$$

Thus we now determine $F_{\text{road}}(t)$.

Step 2: Free-Body Diagram and Force Decomposition Consider the vehicle on an incline of angle α . Using **Theory 1 (Newton's Second Law (Translation))**, the longitudinal resistive force is modeled as

$$F_{\text{road}} = F_{\text{rr}} + F_{\text{grade}} + F_{\text{drag}}. \quad (9.7)$$

Diagram Placeholder (Incline Free-Body Diagram)

Insert a free-body diagram of the vehicle on an incline showing:

- Weight mg acting vertically downward,
- Decomposition into $mg \sin \alpha$ along the slope and $mg \cos \alpha$ normal to the slope,
- Normal force N perpendicular to the surface,
- Positive longitudinal direction $+x$,
- Vehicle velocity v along $+x$.

Step 3: Rolling Resistance from **Theory 14 (Rolling Resistance)**, rolling resistance is modeled as proportional to the normal force:

$$F_{\text{rr}} = C_{\text{rr}} N. \quad (9.8)$$

From the incline free-body diagram, trigonometric decomposition of gravity gives

$$N = mg \cos \alpha. \quad (9.9)$$

Substituting Equation (9.9) into Equation (9.8):

$$F_{\text{rr}} = C_{\text{rr}} mg \cos \alpha. \quad (9.10)$$

Rolling resistance always opposes motion. To enforce directionality, we multiply by the sign of velocity:

$$F_{\text{rr}}(v, \alpha) = C_{\text{rr}} mg \cos \alpha \operatorname{sgn}(v). \quad (9.11)$$

If $v > 0$, the force acts opposite $+x$; if $v < 0$, it acts opposite $-x$.

Step 4: Grade Resistance The longitudinal gravitational component from the free-body diagram is

$$F_{\text{grade}}(\alpha) = mg \sin \alpha. \quad (9.12)$$

This term is positive for an uphill incline ($\alpha > 0$), meaning it resists forward motion.

Step 5: Aerodynamic Drag Following Theory 13 (Air Resistance (Quadratic Drag)), the aerodynamic force magnitude is

$$F_{\text{drag}} = \frac{1}{2} \rho C_d A_f v^2. \quad (9.13)$$

To ensure drag opposes motion, we write

$$F_{\text{drag}}(v) = \frac{1}{2} \rho C_d A_f v^2 \operatorname{sgn}(v). \quad (9.14)$$

Step 6: Total Road Force Substituting Equation (9.11), Equation (9.12), and Equation (9.14) into Equation (9.7) gives

$$\begin{aligned} F_{\text{road}}(t) &= C_{\text{rr}} mg \cos \alpha \operatorname{sgn}(v(t)) \\ &\quad + mg \sin \alpha \\ &\quad + \frac{1}{2} \rho C_d A_f v(t)^2 \operatorname{sgn}(v(t)). \end{aligned} \quad (9.15)$$

Using Equation (9.4), $v(t) = \frac{r_w}{G} \omega_s(t)$, the fully expanded load torque becomes

$$\boxed{\tau_{\text{load}}(t) = \frac{r_w}{G} \left[C_{\text{rr}} mg \cos \alpha \operatorname{sgn}\left(\frac{r_w}{G} \omega_s(t)\right) + mg \sin \alpha + \frac{1}{2} \rho C_d A_f \left(\frac{r_w}{G} \omega_s(t)\right)^2 \operatorname{sgn}\left(\frac{r_w}{G} \omega_s(t)\right) \right]}. \quad (9.16)$$

Equation (9.16) is the explicit resisting torque used in Equation (9.2).

9.5 Inertia Definitions

The rotational equations [Equation \(9.1\)](#)–[Equation \(9.2\)](#) require the effective inertias about each pulley axis.

Primary Pulley Inertia The primary pulley is rigidly connected to the engine. Its effective inertia is therefore the direct sum of

$$I_p = I_{\text{eng}} + I_{\text{prim}}, \quad (9.17)$$

where

- I_{eng} is the engine rotational inertia,
- I_{prim} is the primary CVT pulley inertia.

No reflection is required on the primary side since all components rotate at the same angular velocity ω_p .

Secondary Pulley Inertia All components downstream of the secondary pulley must be expressed as an equivalent inertia about the secondary pulley axis.

Let

- I_{sec} : secondary CVT pulley inertia,
- I_{gb} : gearbox + intermediate shaft inertia,
- I_{wheel} : wheel + hub inertia about the wheel axis,
- m : total vehicle mass,
- r_w : effective wheel rolling radius,
- G : fixed gear ratio between secondary pulley and wheel (defined so that $\omega_s = G\omega_w$).

Vehicle Mass Contribution The vehicle translational kinetic energy is given by [Theory 4 \(Kinetic Energy \(Translational\)\)](#):

$$T_{\text{trans}} = \frac{1}{2}mv^2,$$

with $v = r_w\omega_w$. Expressed in rotational form about the wheel axis:

$$T_{\text{trans}} = \frac{1}{2}(mr_w^2)\omega_w^2.$$

Thus the vehicle mass contributes an effective rotational inertia

$$I_{\text{veh,wheel}} = mr_w^2$$

about the wheel axis.

Reflection to the Secondary Pulley Because $\omega_s = G \omega_w$, we have $\omega_w = \omega_s/G$.

Expressing kinetic energy in terms of ω_s :

$$T = \frac{1}{2} I_{\text{wheel}} \left(\frac{\omega_s}{G} \right)^2 + \frac{1}{2} (mr_w^2) \left(\frac{\omega_s}{G} \right)^2.$$

Therefore inertias at the wheel are reflected to the secondary pulley by

$$I_{\text{reflected}} = \frac{I_{\text{wheel}} + mr_w^2}{G^2}.$$

Total Secondary Inertia The total effective inertia about the secondary pulley becomes

$$I_s = I_{\text{sec}} + I_{\text{gb}} + \frac{I_{\text{wheel}} + mr_w^2}{G^2}. \quad (9.18)$$

Equation [Equation \(9.18\)](#) defines the total inertia resisting angular acceleration of the secondary pulley.

9.6 Ratio Kinematics

The CVT ratio was defined geometrically in [Section 7.9](#) (see [Equation \(7.21\)](#)) as

$$R(t) = \frac{r_{s,\text{eff}}(t)}{r_{p,\text{eff}}(t)}.$$

Under ideal belt transmission, the pulley speeds satisfy

$$\omega_p(t) = R(t) \omega_s(t). \quad (9.19)$$

Differentiating [Equation \(9.19\)](#) with respect to time gives

$$\alpha_p(t) = R(t) \alpha_s(t) + \omega_s(t) \dot{R}(t), \quad (9.20)$$

where $\dot{R}(t) = \frac{dR}{dt}$ was derived previously (see [Equation \(7.49\)](#)).

Equation [Equation \(9.20\)](#) captures the coupling introduced by a time-varying ratio: even if $\alpha_s = 0$, a changing ratio ($\dot{R} \neq 0$) produces primary acceleration.

9.7 Torque Mapping Across the CVT

The belt transmits power from the primary pulley to the secondary pulley. In general,

$$\tau_p(t) \omega_p(t) = \tau_s(t) \omega_s(t) + P_{\text{loss}}(t), \quad (9.21)$$

where $P_{\text{loss}}(t)$ represents internal dissipation. This identity always holds; modeling assumptions determine how P_{loss} is represented.

Following [Assumption 12 \(Ideal CVT Power Transmission\)](#), we set $P_{\text{loss}}(t) = 0$, giving

$$\tau_p(t) \omega_p(t) = \tau_s(t) \omega_s(t). \quad (9.22)$$

Substituting [Equation \(9.19\)](#) yields

$$\tau_s(t) = R(t) \tau_p(t). \quad (9.23)$$

[Equation \(9.23\)](#) is the explicit torque relation required to eliminate τ_s .

9.8 Eliminating Accelerations

We now have four equations ([Equation \(9.1\)](#), [Equation \(9.2\)](#), [Equation \(9.20\)](#), [Equation \(9.23\)](#)) and four unknowns (α_p , α_s , τ_p , τ_s). The goal is to eliminate the accelerations α_p and α_s to obtain a closed-form expression for the coupling torque τ_p in terms of the known quantities τ_{eng} , τ_{load} , R , and $\dot{R}$.

From [Equation \(9.1\)](#),

$$\tau_p = \tau_{\text{eng}} - I_p \alpha_p.$$

Substituting [Equation \(9.20\)](#):

$$\tau_p = \tau_{\text{eng}} - I_p (R \alpha_s + \omega_s \dot{R}). \quad (9.24)$$

From [Equation \(9.2\)](#) and [Equation \(9.23\)](#),

$$\alpha_s = \frac{R \tau_p - \tau_{\text{load}}}{I_s}.$$

Substituting into [Equation \(9.24\)](#):

$$\begin{aligned} \tau_p &= \tau_{\text{eng}} - I_p \left[R \left(\frac{R \tau_p - \tau_{\text{load}}}{I_s} \right) + \omega_s \dot{R} \right] \\ &= \tau_{\text{eng}} - \frac{I_p}{I_s} (R^2 \tau_p - R \tau_{\text{load}}) - I_p \omega_s \dot{R}. \end{aligned} \quad (9.25)$$

Collecting terms:

$$\tau_p \left(1 + \frac{I_p}{I_s} R^2 \right) = \tau_{\text{eng}} + \frac{I_p}{I_s} R \tau_{\text{load}} - I_p \omega_s \dot{R}. \quad (9.26)$$

9.9 Closed-Form Coupling Torque

Solving Equation (9.26):

Coupling Torque (Primary Side)	(9.27)
$\tau_p(t) = \frac{\tau_{\text{eng}}(t) + \frac{I_p}{I_s} R(t) \tau_{\text{load}}(t) - I_p \omega_s(t) \dot{R}(t)}{1 + \frac{I_p}{I_s} R(t)^2}.$	

The coupling torque as seen at the secondary pulley, required by Equation (8.15), then follows from Equation (9.23):

Coupling Torque (Secondary Side)	(9.28)
$\tau_s(t) = R(t) \tau_p(t).$	

9.10 Section Summary

We began with rotational equations of motion for the primary and secondary pulleys. These introduced more unknowns than equations. Closure was obtained by incorporating:

- The geometric ratio definition $R(t)$,
- Its time derivative $\dot{R}(t)$,
- The torque mapping implied by power transfer across the CVT.

The resulting coupling torque Equation (9.27) expresses the torque transmitted through the CVT in terms of the instantaneous ratio and ratio rate. Substitution of Equation (9.28) into Equation (8.15) completes the dynamical coupling required for the axial shift equation Equation (8.35).

10 Traction Limit and Slip Condition

The torque transmission law derived previously assumes ideal no-slip belt traction. Under that assumption, the coupling torque Equation (9.27) (and equivalently Equation (9.28)) represents the torque actually transmitted through the CVT.

In reality, belt-pulley friction is finite. For a given normal clamping force, there exists a maximum tangential traction force that the contact can sustain. If the demanded coupling torque exceeds this traction capacity, slip occurs and the transmitted torque saturates at a friction-limited value.

The objective of this section is to derive a traction inequality of the form

$$|\tau| \leq \tau_{\text{max}}, \quad (10.1)$$

which will later be enforced in place of the ideal coupling law whenever the no-slip solution violates this bound.

10.1 Differential Belt Element and Tangential Force Balance

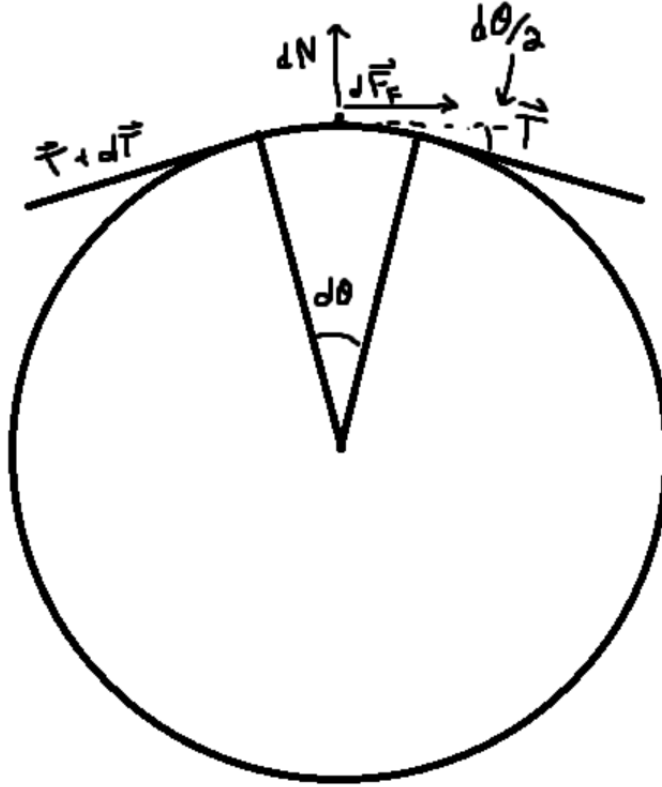


Figure 24: Free-body diagram of a differential belt element spanning $d\theta$.

Consider a differential belt element spanning an infinitesimal wrap angle $d\theta$ on a pulley, as shown in [Figure 24](#). The belt tension entering the element is T , and leaving the element is $T + dT$.

The pulley exerts on the belt element:

- a differential normal force dN (radial),
- a differential friction force dF_f (tangential).

The positive tangential direction is taken along the midpoint tangent of the element, consistent with increasing wrap angle.

Tangential Newton's Second Law Applying [Theory 1 \(Newton's Second Law \(Translation\)\)](#) in the tangential direction:

$$\sum F_t = dm a_t, \quad (10.2)$$

where a_t is the tangential acceleration of the belt material at the contact point.

The tangential components of the end tensions follow directly from geometry:

$$(T)_t = T \cos\left(\frac{d\theta}{2}\right), \quad (10.3)$$

$$(T + dT)_t = (T + dT) \cos\left(\frac{d\theta}{2}\right). \quad (10.4)$$

The left-end tension acts in the positive tangential direction, while the right-end tension acts in the negative tangential direction. Including friction, the tangential force balance becomes

$$T \cos\left(\frac{d\theta}{2}\right) - (T + dT) \cos\left(\frac{d\theta}{2}\right) + dF_f = dm a_t. \quad (10.5)$$

Simplifying the first two terms exactly,

$$-dT \cos\left(\frac{d\theta}{2}\right) + dF_f = dm a_t. \quad (10.6)$$

Rearranging,

$$dT \cos\left(\frac{d\theta}{2}\right) = dF_f - dm a_t. \quad (10.7)$$

Belt mass element From Equation (8.23),

$$dm = \mu_b r_{\text{cm}} d\theta, \quad (10.8)$$

so

$$dm a_t = \mu_b r_{\text{cm}} a_t d\theta. \quad (10.9)$$

Substituting into Equation (10.7),

$$dT \cos\left(\frac{d\theta}{2}\right) = dF_f - \mu_b r_{\text{cm}} a_t d\theta. \quad (10.10)$$

Friction limit and impending slip Theory 10 (Coulomb Friction) states that the friction force magnitude is bounded by the normal force:

$$|dF_f| \leq \mu dN. \quad (10.11)$$

The maximum transmissible tension increase occurs when friction is fully mobilized in the direction that increases tension (impending slip):

$$dF_f = +\mu dN. \quad (10.12)$$

Substituting Equation (10.12) into Equation (10.10) gives the maximum local tension change:

$$dT_{\max} = \frac{\mu dN - \mu_b r_{\text{cm}} a_t d\theta}{\cos\left(\frac{d\theta}{2}\right)}. \quad (10.13)$$

10.2 Integration Across the Wrap

Define the tight-slack tension difference

$$\Delta T = T_{\text{tight}} - T_{\text{slack}} = \int_0^\phi dT. \quad (10.14)$$

The maximum admissible tension difference is therefore

$$\Delta T_{\max} = \int_0^\phi \frac{\mu dN - \mu_b r_{\text{cm}} a_t d\theta}{\cos\left(\frac{d\theta}{2}\right)}. \quad (10.15)$$

In the differential limit $d\theta \rightarrow 0$, $\cos(d\theta/2) \rightarrow 1$, yielding

$$\Delta T_{\max} = \int_0^\phi \mu dN - \int_0^\phi \mu_b r_{\text{cm}} a_t d\theta. \quad (10.16)$$

Define total normal force over the wrap:

$$N_\phi = \int_0^\phi dN. \quad (10.17)$$

Assuming r_{cm} and a_t are approximately constant over the wrap,

$$\Delta T_{\max} = \mu N_\phi - \mu_b r_{\text{cm}} a_t \phi. \quad (10.18)$$

10.3 Torque Capacity Inequality

The transmitted torque magnitude is

$$|\tau| = r_{\text{eff}} |\Delta T|. \quad (10.19)$$

Substituting Equation (10.18) gives the traction limit

$$|\tau| \leq r_{\text{eff}} (\mu N_\phi - \mu_b r_{\text{cm}} a_t \phi). \quad (10.20)$$

Equation (10.20) provides the general traction-limited torque bound. However, two quantities remain to be defined: the normal force per unit arc length N_ϕ and the tangential acceleration a_t . In the next subsection, a_t will be expressed explicitly in terms of pulley angular acceleration and radial rate.

10.4 Tangential Acceleration of a Belt Element on a Circular Wrap

This subsection derives the tangential acceleration term required in the local traction inequality in §??. In that inequality we apply Newton's second law *along the direction of belt motion*, so we need an explicit expression for the belt element's tangential acceleration a_t .

Goal For a belt element of differential mass dm moving along the wrap, the tangential force balance uses

$$\sum F_t = dm a_t,$$

so our goal is to compute a_t from the belt kinematics when both *angular speed* and *wrap radius* may vary:

$$\omega(t) = \dot{\theta}(t), \quad \alpha(t) = \dot{\omega}(t), \quad r(t), \quad \dot{r}(t).$$

Coordinate system and unit vectors Because we care about forces *along the belt*, we choose a basis aligned with the belt's local geometry:

- $\mathbf{e}_r(t)$ points radially outward from the pulley center O to the belt point P ,
- $\mathbf{e}_\theta(t)$ is the unit tangent, obtained by rotating $\mathbf{e}_r$ by $+90^\circ$ in the direction of increasing θ .

These are not arbitrary symbols: they are *the* directions of radial and tangential motion on a circle, and the belt's local tangential direction is exactly $\mathbf{e}_\theta$.

Placeholder: polar basis at the wrap. Draw pulley center O and belt point P at radius r and angle θ . Show $\mathbf{e}_r$ pointing from O to P and $\mathbf{e}_\theta$ perpendicular to $\mathbf{e}_r$ in the direction of belt motion (increasing θ). Label belt tangential direction as $\mathbf{e}_\theta$.

Figure 25: Definition of the radial and tangential unit vectors.

Work backwards: to get a_t , compute $\mathbf{a}$ and take its tangential component Tangential acceleration is the component of the acceleration vector along $\mathbf{e}_\theta$:

$$a_t \equiv \mathbf{a} \cdot \mathbf{e}_\theta. \quad (10.21)$$

So we will compute $\mathbf{a} = \dot{\mathbf{v}}$, then read off the $\mathbf{e}_\theta$ component.

Step 1: position and velocity The position of P from the center is

$$\mathbf{r}(t) = r(t) \mathbf{e}_r(t). \quad (10.22)$$

Differentiate **Equation (10.22)** using the product rule:

$$\mathbf{v} = \dot{\mathbf{r}} = \dot{r} \mathbf{e}_r + r \dot{\mathbf{e}}_r. \quad (10.23)$$

This shows why we must track how the unit vectors change with time: even if r were constant, $\mathbf{e}_r$ rotates as the belt moves around the pulley.

Step 2: derive $\dot{\mathbf{e}}_r$ and $\dot{\mathbf{e}}_\theta$ (no assumptions) Over a small change $d\theta$, the unit vectors rotate rigidly by the same angle. The derivative with respect to angle is therefore:

$$\frac{d\mathbf{e}_r}{d\theta} = \mathbf{e}_\theta, \quad \frac{d\mathbf{e}_\theta}{d\theta} = -\mathbf{e}_r. \quad (10.24)$$

(Geometric justification: increasing θ rotates $\mathbf{e}_r$ toward the tangent; increasing θ rotates the tangent toward $-\mathbf{e}_r$.)

Convert angle-derivatives to time-derivatives with the chain rule $d(\cdot)/dt = (d(\cdot)/d\theta)\dot{\theta}$ and $\omega = \dot{\theta}$:

$$\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r. \quad (10.25)$$

Step 3: velocity in radial–tangential form Substitute $\dot{\mathbf{e}}_r$ from Equation (10.25) into Equation (10.23):

$$\mathbf{v} = \dot{r} \mathbf{e}_r + r\omega \mathbf{e}_\theta. \quad (10.26)$$

This is already intuitive: $\dot{r}$ is the radial speed, and $r\omega$ is the tangential speed.

Step 4: acceleration and its tangential component Differentiate Equation (10.26):

$$\begin{aligned} \mathbf{a} = \dot{\mathbf{v}} &= \frac{d}{dt}(\dot{r} \mathbf{e}_r) + \frac{d}{dt}(r\omega \mathbf{e}_\theta) \\ &= \ddot{r} \mathbf{e}_r + \dot{r} \dot{\mathbf{e}}_r + (\dot{r}\omega + r\alpha) \mathbf{e}_\theta + r\omega \dot{\mathbf{e}}_\theta. \end{aligned} \quad (10.27)$$

Now substitute the unit-vector time derivatives from Equation (10.25):

$$\dot{\mathbf{e}}_r = \omega \mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\omega \mathbf{e}_r,$$

and collect like directions:

$$\begin{aligned} \mathbf{a} &= \ddot{r} \mathbf{e}_r + \dot{r}(\omega \mathbf{e}_\theta) + (\dot{r}\omega + r\alpha) \mathbf{e}_\theta + r\omega(-\omega \mathbf{e}_r) \\ &= (\ddot{r} - r\omega^2) \mathbf{e}_r + (r\alpha + 2\dot{r}\omega) \mathbf{e}_\theta. \end{aligned} \quad (10.28)$$

Finally, by the definition Equation (10.21), the tangential acceleration is the $\mathbf{e}_\theta$ component:

$$a_t = r\alpha + 2\dot{r}\omega. \quad (10.29)$$

Where the factor of 2 comes from The two $\dot{r}\omega$ contributions in Equation (10.28) come from *two distinct effects*:

- $\dot{r} \dot{\mathbf{e}}_r$: even the *radial* velocity $\dot{r} \mathbf{e}_r$ gains a tangential contribution because $\mathbf{e}_r$ rotates at rate ω ;
- $\frac{d}{dt}(r\omega)$: the tangential speed $r\omega$ changes if r changes.

Adding these yields the total $2\dot{r}\omega$ term.

10.5 Normal Force from Axial Clamping

The local traction inequality derived in ?? depends on the normal load over the wrap,

$$N_\phi \equiv \int_0^\phi dN,$$

since the available friction is μdN at each differential contact element.

In the CVT, this normal load is not generated implicitly by belt tension (as in the classical capstan problem). Instead, it is imposed by the pulley *axial clamping force* derived in Section 8. We now map axial clamp into an equivalent total normal load over the wrap.

Geometry: why a factor of $2 \tan \beta$ appears Recall the conical kinematics established earlier for opposing sheaves (Section 7.3), where axial closure beyond the deadzone produces a radial change

$$\Delta r = \frac{\Delta s}{2 \tan \beta} \quad \implies \quad \frac{dr}{ds} = \frac{1}{2 \tan \beta}.$$

This is the key geometric ratio: a given axial displacement ds produces a smaller radial displacement dr by the factor $2 \tan \beta$, because the closure is shared between the two opposing faces.

To convert forces consistently, we use virtual work (equivalently, power balance in the shift direction): the incremental work done by the axial clamp must equal the incremental work associated with the corresponding radial normal loading,

$$F_{\text{ax}} ds = F_{\text{rad}} dr.$$

Substituting $dr = (dr/ds) ds$ gives

$$\begin{aligned} F_{\text{ax}} ds &= F_{\text{rad}} \left(\frac{dr}{ds} \right) ds \\ \implies F_{\text{rad}} &= \frac{F_{\text{ax}}}{dr/ds} = F_{\text{ax}} (2 \tan \beta). \end{aligned} \tag{10.30}$$

Thus the total radial normal load supplied by a given axial clamp force is

$$F_{\text{rad}} = 2 \tan \beta F_{\text{ax}}. \tag{10.31}$$

Identification with wrap-normal load N_ϕ Under the uniform distribution assumption (discussed previously), we take the total normal load over the wrap to be equal to the total radial normal load supplied by clamping:

$$N_\phi = F_{\text{rad}} = 2 \tan \beta F_{\text{ax}}. \tag{10.32}$$

Equivalently, the normal force per unit wrap angle is constant:

$$\frac{dN}{d\theta} = \frac{N_\phi}{\phi} \quad \implies \quad dN = \frac{N_\phi}{\phi} d\theta. \tag{10.33}$$

Substituting Equation (10.32) into the integrated traction bound (from ??) yields the clamp-limited torque capacity form

$$|\tau| \leq r_{\text{eff}} [\mu N_\phi - \mu_b r_{\text{cm}} a_t \phi] = r_{\text{eff}} [\mu (2 \tan \beta F_{\text{ax}}) - \mu_b r_{\text{cm}} a_t \phi]. \quad (10.34)$$

Remark (Why this differs from the classical capstan exponential)

In the classical capstan problem, the normal load is generated *by the belt tension* itself (a larger tension implies a proportionally larger contact normal), which leads to the exponential relation $T_1/T_2 = e^{\mu\phi}$.

Here, the normal load is imposed externally by the clamping mechanism: N_ϕ is determined by F_{ax} through Equation (10.32). Therefore, the available traction scales *linearly* with clamping force, not exponentially with belt tension. Centrifugal loading and ratio change enter additionally through the inertial term in Equation (10.34).

10.6 Section Conclusion and Transition

This section derived a traction-limited torque bound for a single pulley.

The result is

$$|\tau| \leq r_{\text{eff}} [\mu (2 \tan \beta F_{\text{ax}}) - \mu_b r_{\text{cm}} a_t \phi], \quad (10.35)$$

with

$$a_t = r_{\text{cm}} \alpha + 2\dot{r}_{\text{cm}} \omega.$$

Equation Equation (10.35) gives the maximum torque that can be transmitted at the current operating state. It depends only on:

- axial clamping force F_{ax} ,
- cone angle β ,
- wrap angle ϕ ,
- belt mass and kinematics.

No exponential amplification occurs because the normal force is externally imposed and bounded.

The ideal coupling torque derived earlier remains valid only when it satisfies Equation (10.35). If the demanded torque exceeds this bound, the belt cannot sustain static friction and slip must occur.

The next section defines the stick and slip regimes and modifies the governing equations accordingly.

11 Slip Regimes and Traction-Limited Coupling Torque

Placeholder: Slip intuition diagram.

Sketch two pulleys with effective radii $r_{p,\text{eff}}$, $r_{s,\text{eff}}$ and angular speeds ω_p, ω_s . Show the two tangential surface speeds $v_p = r_{p,\text{eff}}\omega_p$ and $v_s = r_{s,\text{eff}}\omega_s$ at the belt contact, and define $v_{\text{rel}} = v_p - v_s$. Annotate: $v_{\text{rel}} = 0$ (stick) vs $v_{\text{rel}} \neq 0$ (slip), and indicate the friction torque direction opposing v_{rel} .

Figure 26: Conceptual definition of slip via contact-speed mismatch.

11.1 Working Backwards From the Goal

The goal of this section is to define, at every instant, a *physically admissible* coupling torque that the belt can transmit between the pulleys. Earlier, we derived:

- a no-slip (rigid-traction) relationship that specifies how torque transfers between the pulleys when the belt adheres (“stick”), and
- a traction capacity bound for each pulley of the form $|\tau| \leq \tau_{\text{max},p}$ and $|\tau| \leq \tau_{\text{max},s}$.

These two results must be combined into a single rule that:

1. *never* violates either capacity bound (even before slip begins),
2. reduces to the earlier no-slip torque transfer when feasible, and
3. produces a slip torque that opposes relative motion when the no-slip demand is infeasible.

To do this cleanly, we proceed in the order that matches physical intuition:

(i) *define how we detect slip* $\rightarrow$ (ii) *define the torque limits that force entry into slip* $\rightarrow$ (iii) *define the torque law during slip* $\rightarrow$ (iv) *define how we exit slip*.

11.2 Slip Metric: How We Know We Are Slipping

Slip in a belt CVT is most naturally defined at the belt–pulley contact: if the belt adheres to both pulleys, then the tangential surface speeds at the contact must match. Using the effective contact radii, define

$$v_p \triangleq r_{p,\text{eff}} \omega_p, \quad v_s \triangleq r_{s,\text{eff}} \omega_s, \quad (11.1)$$

and the relative contact-speed mismatch

$$v_{\text{rel}} \triangleq v_p - v_s = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s. \quad (11.2)$$

Stick condition (no slip):

$$v_{\text{rel}} = 0. \quad (11.3)$$

Slip condition:

$$v_{\text{rel}} \neq 0. \quad (11.4)$$

Equation [Equation \(11.2\)](#) is therefore the diagnostic: it is the quantity we monitor to determine whether the belt is adhering (stick) or sliding (slip). In a perfect rigid-traction model, the dynamics

would enforce Equation (11.3) exactly. In a traction-limited model, Equation (11.3) can only be maintained if the required torques remain within the available traction capacity.

11.3 Traction Capacity: How We Enter Slip

From the traction analysis of the previous section, each pulley has a maximum transmissible torque magnitude at the current operating state:

$$|\tau_p| \leq \tau_{\max,p}(\cdot), \quad |\tau_s| \leq \tau_{\max,s}(\cdot), \quad (11.5)$$

where $\tau_{\max,p}$ and $\tau_{\max,s}$ depend on the current state (clamping force, cone angle, wrap angle, belt kinematics, etc.) through Equation (10.35) applied to the primary and secondary, respectively.

Key physical point: these bounds apply *always*. Even before any slip is observed, the belt–pulley interface cannot transmit a torque exceeding its instantaneous capacity. Therefore, if the no-slip model (rigid traction) would *demand* a torque larger than the bound in Equation (11.5), the constraint $v_{\text{rel}} = 0$ can no longer be maintained by static friction.

In practice, torque transfer is constrained by the *most limiting* interface. We therefore define a single admissible coupling capacity

$$\tau_{\max} \triangleq \min(\tau_{\max,p}, \tau_{\max,s}). \quad (11.6)$$

Intuition for entering slip. Suppose the system is in stick so that $v_{\text{rel}} = 0$ initially. External conditions (engine torque rise, vehicle load increase, ratio change reducing effective radius leverage, reduced clamping, etc.) may increase the torque required to preserve $v_{\text{rel}} = 0$. As soon as the required coupling torque exceeds $\tau_{\max}$, the transmitted torque cannot increase further. The coupling torque saturates at the traction limit, while the remaining unbalanced torque accelerates one pulley relative to the other, causing Equation (11.3) to break and v_{rel} to grow in magnitude. This is the mechanism by which the model *falls into* a slip regime.

11.4 Coupling Torque Law: Stick When Feasible, Slip When Necessary

Earlier sections already specify how torques transfer *when not slipping*: the dynamics and kinematics provide a “desired” (no-slip) coupling torque that would maintain $v_{\text{rel}} = 0$. We denote this no-slip demand by

$$\tau_{\text{ns}} \triangleq (\text{the coupling torque predicted by the no-slip torque-transfer model}). \quad (11.7)$$

This quantity is not a new model; it is simply a name for “whatever torque the earlier no-slip equations imply” at the current state.

However, the belt cannot transmit an arbitrarily large torque: it must also satisfy Equation (11.6). Therefore, even in stick, the admissible coupling torque is at most $\tau_{\max}$ in magnitude.

(A) Feasible no-slip torque (stick). If the no-slip demand is within capacity,

$$|\tau_{\text{ns}}| \leq \tau_{\max}, \quad (11.8)$$

then static friction can support the demand and the belt remains in stick:

$$\tau_c = \tau_{\text{ns}}, \quad v_{\text{rel}} = 0. \quad (11.9)$$

(B) Infeasible no-slip torque (entering slip). If the no-slip demand exceeds capacity,

$$|\tau_{\text{ns}}| > \tau_{\text{max}}, \quad (11.10)$$

then the coupling torque must saturate and the mismatch v_{rel} can no longer remain zero. In slip, the transmitted torque is friction-limited and its direction must oppose the relative motion. A smooth Coulomb-like slip law is

$$\tau_{\text{slip}} = \tau_{\text{max}} \tanh\left(\frac{v_{\text{rel}}}{v_{\text{smooth}}}\right), \quad (11.11)$$

where $v_{\text{smooth}} > 0$ controls the transition sharpness near $v_{\text{rel}} = 0$. For $|v_{\text{rel}}| \gg v_{\text{smooth}}$, this approaches $\tau_{\text{max}} \text{sign}(v_{\text{rel}})$.

Sign convention note. Equation [Equation \(11.11\)](#) is written so that the slip torque changes sign with v_{rel} . In the equations of motion, τ_c must be applied with opposite sign on the two pulleys (equal-and-opposite action), so that the net effect is always to *reduce* the mismatch and dissipate energy during slip. (If your earlier rotational EOM use the opposite sign convention for τ_c , the tanh argument should be negated accordingly.)

(C) Smooth unified law (recommended for numerics). Rather than hard switching, we define a blending factor $\alpha \in [0, 1]$ that increases from stick-like to slip-like behavior as the relative speed grows:

$$\alpha(v_{\text{rel}}) = \text{clip}\left(\frac{|v_{\text{rel}}|}{v_{\text{blend}}}, 0, 1\right), \quad (11.12)$$

where $v_{\text{blend}} > 0$ is a chosen blending scale. We also define the capacity-saturated no-slip torque

$$\tau_{\text{ns,sat}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\text{max}}, +\tau_{\text{max}}). \quad (11.13)$$

The final coupling torque is then

$$\tau_c = (1 - \alpha) \tau_{\text{ns,sat}} + \alpha \tau_{\text{slip}}, \quad (11.14)$$

with τ_{slip} given by [Equation \(11.11\)](#).

This construction encodes all desired behaviors:

- **Capacity is never violated:** both $\tau_{\text{ns,sat}}$ and τ_{slip} are bounded by τ_{max} , hence so is τ_c .
- **Stick when feasible:** if $v_{\text{rel}} \approx 0$, then $\alpha \approx 0$ and $\tau_c \approx \tau_{\text{ns,sat}}$ (equal to τ_{ns} if feasible).
- **Slip when necessary:** if mismatch grows, $\alpha \rightarrow 1$ and $\tau_c \rightarrow \tau_{\text{slip}}$, i.e. friction-limited sliding torque.

11.5 How Slip Reduces and Ends: Exiting the Slip Regime

Once the coupling torque saturates, the remainder of the external torque imbalance accelerates the pulleys differently, which produces a nonzero v_{rel} through [Equation \(11.2\)](#). During slip, the transmitted torque magnitude cannot exceed τ_{max} , but its *direction* always opposes the mismatch via [Equation \(11.11\)](#). As a result, the slip torque acts to reduce $|v_{\text{rel}}|$ whenever the external conditions permit.

Intuition for exiting slip. Slip persists as long as the system would require more than τ_{max} to enforce $v_{\text{rel}} = 0$. If operating conditions change (e.g., demanded torque falls, clamping increases,

wrap angle increases, ratio shifts into a more favorable radius, or the inertial term reduces), then $\tau_{\max}$ may rise and/or the required no-slip demand τ_{ns} may fall. When the inequality becomes feasible again,

$$|\tau_{\text{ns}}| \leq \tau_{\max}, \quad (11.15)$$

static traction can re-establish adherence and the mismatch decays to zero. In the smooth blended law [Equation \(11.14\)](#), this corresponds to $v_{\text{rel}} \rightarrow 0$, so that $\alpha \rightarrow 0$ and $\tau_c \rightarrow \tau_{\text{ns}}$ continuously.

11.6 Implementation Summary (Matches the Simulation Rule)

For convenience, the continuous-time model above corresponds directly to the following computation at each time step:

1. compute $v_{\text{rel}} = r_{p,\text{eff}}\omega_p - r_{s,\text{eff}}\omega_s$,
2. compute capacities $\tau_{\max,p}$ and $\tau_{\max,s}$, then $\tau_{\max} = \min(\tau_{\max,p}, \tau_{\max,s})$,
3. compute the no-slip demand τ_{ns} from the earlier torque-transfer equations,
4. form $\tau_{\text{ns,sat}} = \text{clip}(\tau_{\text{ns}}, -\tau_{\max}, +\tau_{\max})$,
5. form $\tau_{\text{slip}} = \tau_{\max} \tanh(v_{\text{rel}}/v_{\text{smooth}})$,
6. blend using $\alpha = \text{clip}(|v_{\text{rel}}|/v_{\text{blend}}, 0, 1)$: $\tau_c = (1 - \alpha)\tau_{\text{ns,sat}} + \alpha\tau_{\text{slip}}$.

This yields a coupling torque that (i) respects traction limits at both pulleys, (ii) reproduces the earlier no-slip behavior when feasible, and (iii) transitions smoothly into a Coulomb-like slip regime when necessary.

A Parameter Tables

B Belt Length Small-Angle Approximation

This section quantifies the error introduced by the small-angle approximation commonly used in simplified CVT kinematic models.

B.1 Standard Approximation

Many treatments assume

$$\frac{r_s - r_p}{C} \ll 1,$$

which leads to $\lambda \approx 0$ and therefore

B.2 Error Analysis

C Common Ramp Profile Designs

This section catalogs several common ramp profile parameterizations used in CVT primary pulley actuation.

C.1 Linear Ramp

The simplest ramp profile is a linear relationship:

$$r_f(s) = r_{f,0} + k_r \cdot s,$$

where k_r is the ramp slope (constant) and $r_{f,0}$ is the initial flyweight radius.

The corresponding ramp angle is constant:

$$\alpha_p(s) = \tan^{-1}(k_r).$$

Characteristics:

- Uniform force multiplication across the shift range
- Simple to manufacture and analyze
- May not provide optimal shift feel or calibration flexibility

C.2 Quadratic Ramp

C.3 Piecewise-Linear Ramp

C.4 Custom Profiles from Lookup Tables

D Numerical Methods Summary

This section summarizes the numerical solution methods used throughout this work.

D.1 Root Finding for Belt-Length Constraint

The belt-length constraint [Equation \(7.11\)](#) is solved using Brent's method, a robust bracketed root-finding algorithm combining bisection, secant, and inverse quadratic interpolation.

D.2 ODE Integration for Dynamic Simulation

E Nomenclature and Symbol Reference

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